

Consumer hardware, loyalty & public services

LUCKNOW, A SATURDAY IN APRIL, THE WEEK BEFORE POLLING DAY

Ananya, 21, will vote for the first time on Thursday. She has spent ₹25,000 on AirPods and an Apple Watch this year, but she hasn't checked which booth she votes at, and she isn't sure her name is on the list. In the kitchen her mother pays for groceries on BigBasket and the app tells her she has 1,240 NeuCoins she never noticed. Her grandfather is stopped at a traffic check without his driving licence; he opens DigiLocker and shows it on his phone, and the constable accepts it. The city's turnout last time was 55%.

Try it: Ananya will spend ₹25,000 on gadgets but may not spend 20 minutes voting. What would you build to change that, and how would you know it worked? Think for five minutes before reading on.

Three businesses that run on one habit

A smart speaker, a loyalty programme and a government app look unrelated. They share one job: get a person to come back and do a small thing again, long after the first sale or the first download. Apple sells a speaker once but earns every month from the music, storage and warranty plans that ride on it. A loyalty programme gives away points today so that the customer picks Tanishq over another jeweller next year. A voter app is worth nothing if it is downloaded and never opened in voting week. In all three, the first transaction is the easy part, and the product is judged on what happens after it.

\$109 bn

Apple's services revenue in one year, riding on devices it already sold (FY2025)

67.6 crore

DigiLocker users; 950 crore+ documents issued by the issuers (PIB, Mar 2026)

₹4,974 Cr

Tata Digital's loss in FY26, on ₹35,990 crore of revenue (Sep 2026 filings)

65.8%

Lok Sabha turnout in 2024, down from 67.4% in 2019 (Election Commission)

Consumer hardware: who does what, and who pays whom

Follow one HomePod mini from the factory to a living room in Lucknow. A contract manufacturer assembles it from chips, a speaker driver, a microphone array and a plastic mesh body. Apple ships it to India, pays import duty, and sells it through its own stores, its website and resellers such as Croma, Imagine and Amazon. The buyer pays the price on the box, called the **MRP** (maximum retail price, which in India includes GST). Of ₹15,900, the government takes ₹2,425 as GST, the shop and distributor keep a margin, the parts and assembly cost several thousand rupees, and Apple keeps what is left (Figure 9.1).

One HomePod mini sold in India: where the ₹15,900 goes (illustrative split)

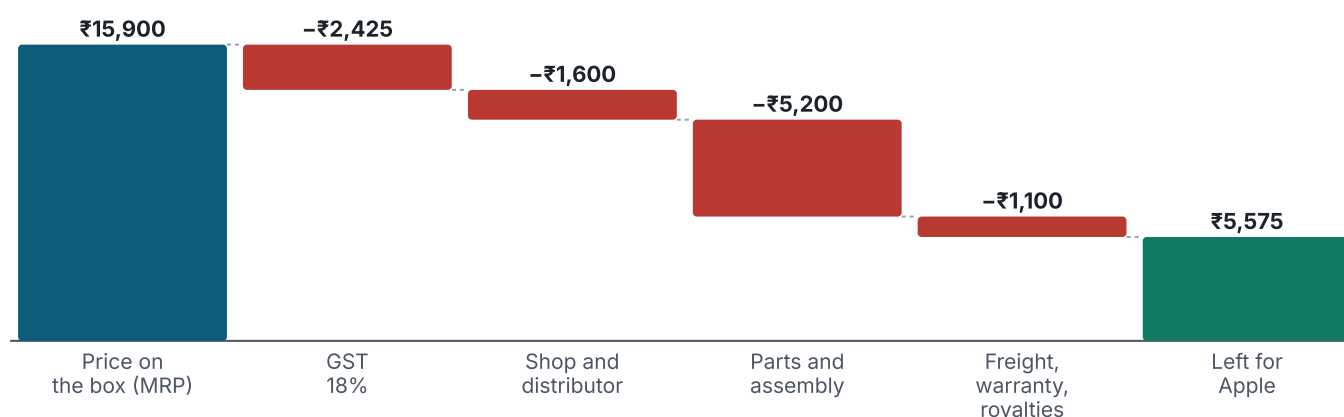


Figure 9.1 · Where ₹15,900 of a smart speaker goes. The tax and the shop take about a quarter before Apple sees anything. The part Apple keeps is large for a gadget, and that is by choice: it prices for the people who already own an iPhone. *MRP from Apple's India store (Sep 2026); GST at 18%; the other splits are illustrative, not Apple's figures.*

Two ways to make money from a device. Apple keeps a high margin on the device itself and then sells services on top. Amazon did the opposite with the Echo: it sold the speaker near cost, often under ₹3,000 in a sale, because the speaker was a way to put Alexa in the home, and Alexa was meant to earn through shopping ("Alexa, reorder atta"). The first model works when people will pay for the brand and the device is good enough to stand alone. The second works only if the thing the device sells actually gets used. Voice shopping never took off, and reports in 2022 put the losses of Amazon's devices unit at billions of dollars a year. The lesson for a PM: a device sold at a loss is a bet on what comes after, so measure what comes after.

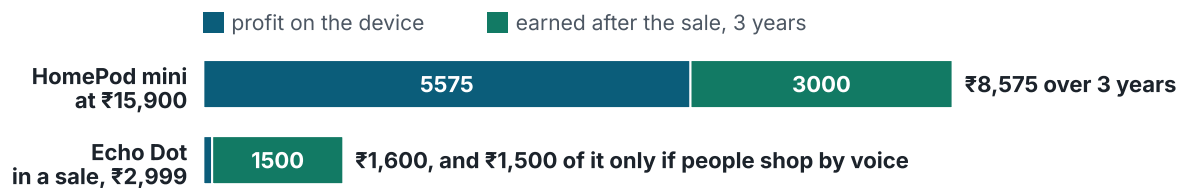


Figure 9.2 • Two ways to earn from a device, per unit. Apple keeps a large profit on the speaker and adds services on top. Amazon sold the Echo near cost and counted on shopping by voice, which is the part that never arrived. *Illustrative amounts, not Apple's or Amazon's reported figures.*

The ecosystem. Apple's real product is the set of devices working together (Figure 9.3). AirPods switch from the iPhone to the Mac without pairing again. An Apple Watch works only with an iPhone. A HomePod plays what the phone was playing. Each extra device makes the others more useful, and together they make leaving expensive: moving to Android means replacing the watch and losing half of what the earbuds do. The number that tracks this is the **attach rate**: the share of owners of one device who also buy another (for example, 35 of every 100 iPhone owners buying AirPods). The money earned after the sale is **services revenue per device**: iCloud storage, Apple Music, AppleCare and a cut of App Store sales, paid every month.

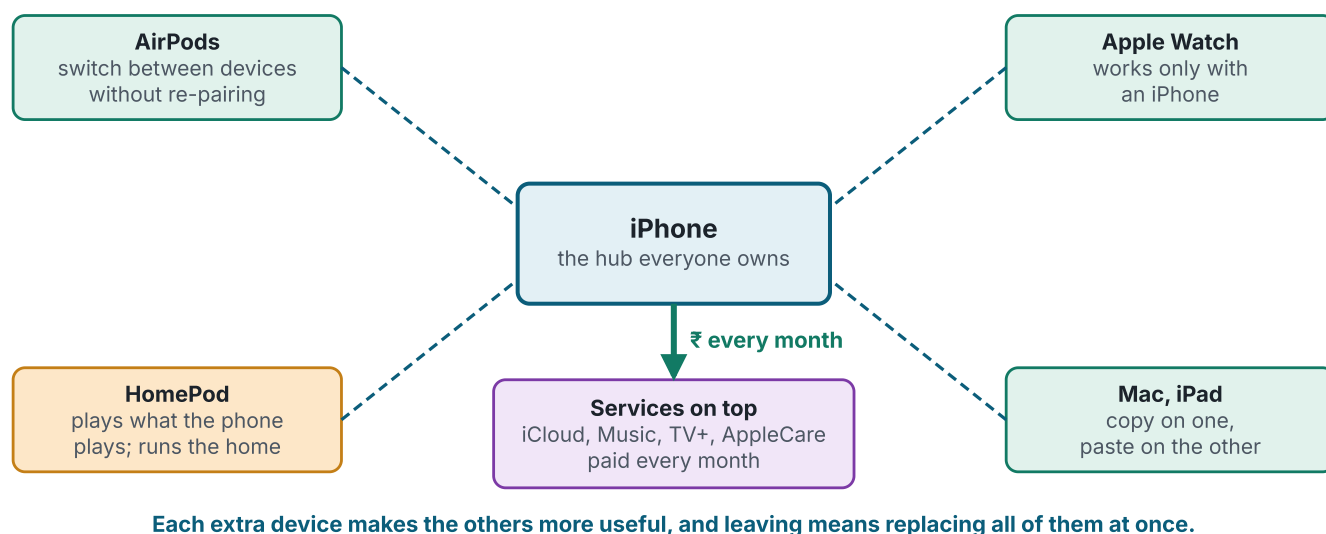
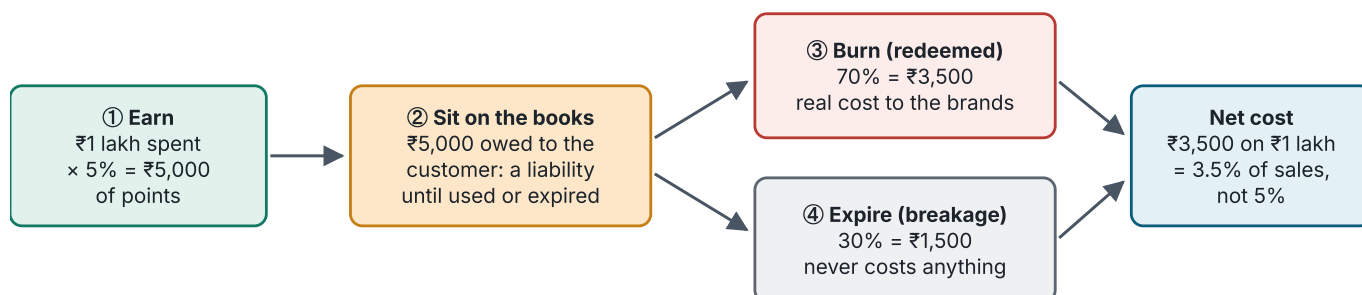


Figure 9.3 • The ecosystem web. The iPhone sits in the middle; each device around it adds something that only works with the others. Services are paid monthly by the people inside the web, which is why Apple can afford to price devices for the people already in it. *Author's diagram; features as of 2026.*

Loyalty: how a point is born, and who pays for it

A loyalty point is a promise: “spend with us, and we owe you something later”. Say Priya spends ₹1 lakh a year across Tata brands and earns 5% back, ₹5,000 of points. That ₹5,000 is the **earn rate** at work. The moment the points are given, the company owes them to her, so accountants record them as a **liability** (money owed, sitting on the balance sheet). Some points are later spent, the **burn** or redemption; some expire unused, which is called **breakage**. If 70% are redeemed, the real cost is ₹3,500, or 3.5% of what she spent, not 5% (Figure 9.4).

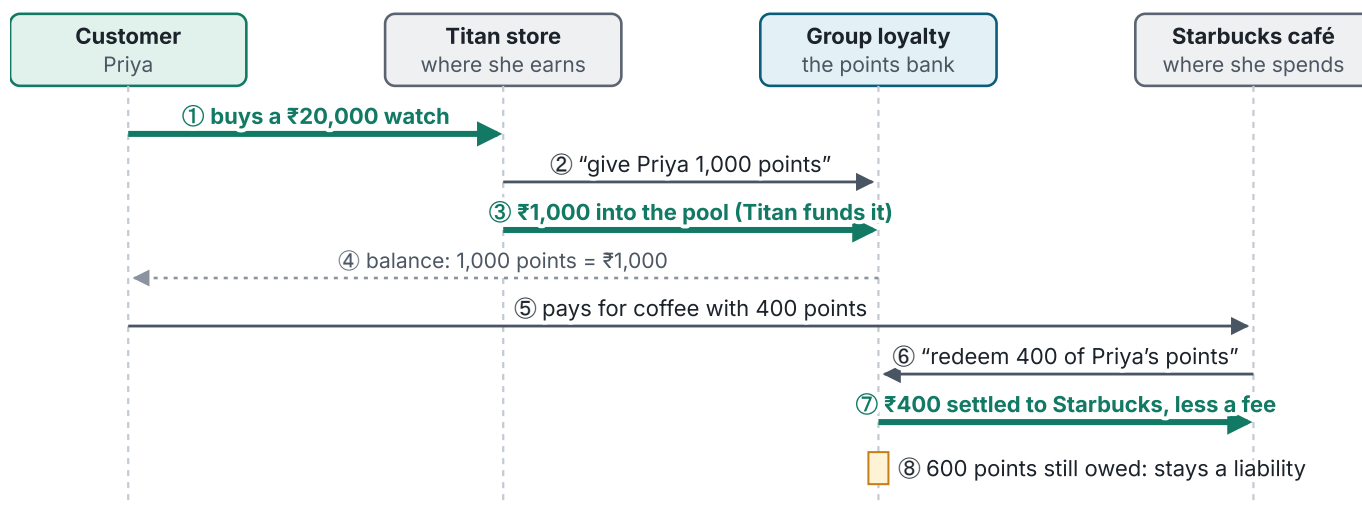


A programme that is too stingy (little to spend points on) raises breakage and looks cheap on paper, but customers notice that their points are worthless and stop caring.

Figure 9.4 • The life of a loyalty point. A point is a cost only when it is spent. Breakage makes a programme look cheaper, but a programme that relies on it is one where customers don't value the points. *Illustrative rates; Tata Neu's NeuCoin is worth ₹1 (Tata Neu, 2026).*

The hard part is when points cross brands. In a single-brand programme the shop that gives the point also takes it back. In a group programme, Priya earns points at Titan and spends them at a Starbucks café. Starbucks gave away ₹400 of coffee and must be paid, so somebody runs a clearing account: the brand that issues a point pays money into a pool, and the

brand that accepts it is paid from the pool, often less a small fee (Figure 9.5). Airline programmes work the same way: a bank buys miles from the airline to give to its card users, which is why co-branded credit cards are the most profitable part of many loyalty programmes. Tata Neu's card with HDFC Bank had passed 20 lakh cards by March 2025.



The brand that issues the point pays for it; the brand that accepts it gets paid. Without that rule no brand will join.

Figure 9.5 · Who pays when a point crosses brands. Titan funds the point when it is earned (③); Starbucks is paid when it is spent (⑦); the unspent part stays on the books as money owed (⑧). *Author's diagram; amounts and the fee are illustrative.*

Public digital services: the state as a product builder

India built a set of public systems, often called **digital public infrastructure** (DPI): shared systems built once by the state that any bank, company or department can plug into (Figure 9.6). Aadhaar lets anyone prove who they are. UPI moves money between any two bank accounts. DigiLocker holds documents fetched straight from the body that issued them, such as the transport department's driving licence or CBSE's mark sheet. ONDC lets any shopping app show any seller's goods. Private companies then build apps on top and compete for users.

Commerce	ONDC (2022)	an open network any buyer app and seller app can join
Consent to share data	Account Aggregator (2021)	you approve which bank data a lender may read
Documents	DigiLocker (2015)	67.6 crore users; 950 crore+ documents issued
Payments	UPI (2016)	about 2,000 crore payments a month
Identity	Aadhaar (2010)	a number anyone can verify with a fingerprint or OTP

Each layer is built once by the state (or a non-profit), and any bank, app or shop can plug in.

Private companies compete on the apps built on top: PhonePe on UPI, Magicpin on ONDC, lenders on Account Aggregator.

Figure 9.6 · India's public digital layers. The layers are shared roads; companies build the vehicles. A public app succeeds when the layer is trusted and the apps on it are good enough that people choose them. *PIB (Mar 2026) for DigiLocker; NPCI for UPI (2026, rounded); launch years from the programmes.*

Success is measured differently. A government doesn't earn from DigiLocker. It succeeds if people can use it (reach), if officials and citizens accept what it shows (trust), and if a task becomes cheaper for the citizen (fewer trips to an office, fewer photocopies). The trap is to count downloads. A voter app downloaded by lakhs during a campaign may be opened by very few in voting week, and fewer still vote because of it (Figure 9.7). The people who install a voter app are also the keen ones who would have voted anyway, which makes "app users vote more" a misleading number.

A public app, 100 people who downloaded it (illustrative)

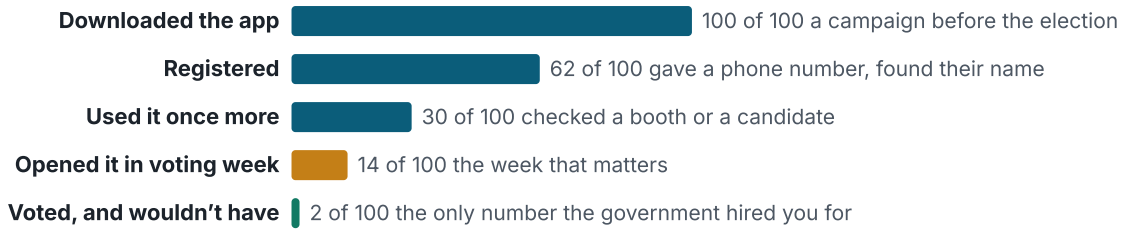


Figure 9.7 · From download to an extra vote. Each step loses people. Only the last bar is what a turnout campaign is for, and it can only be measured by comparing with people who didn't get the nudge. *Illustrative numbers.*

The metrics that matter

Metric	What it means, in one line	Example
Attach rate	Share of owners of one device who also own another	35 of 100 iPhone owners have AirPods = 35%
Active installed base	Devices in use in a period, not devices ever sold	Apple reported over 2.35 billion active devices (Jan 2025)
Services revenue per device	What each active device earns a year after its sale	₹6,000 a year of iCloud, music and AppleCare (illustrative)
Earn rate	Points given per ₹100 spent, at the value of a point	5 NeuCoins per ₹100 = 5%
Burn (redemption) rate	Share of issued points that are spent	₹3,500 of ₹5,000 redeemed = 70%
Breakage	Share of points that expire unused	₹1,500 of ₹5,000 = 30%
Cross-brand shoppers	Customers buying from two or more brands in the group	18 of 100 members shop at two brands (illustrative)
Enrolment vs active use	People who signed up vs people who used it in the period that matters	62 registered, 14 opened it in voting week
Turnout	Votes cast ÷ people on the electoral roll	65.8% in the 2024 Lok Sabha election

Players and numbers

Consumer hardware. Apple dominates the premium end in India (phones over ₹50,000), while Samsung, Xiaomi, Vivo and Oppo sell most phones overall. In smart speakers Amazon's Echo sells most units, helped by frequent deep discounts; Google's Nest Mini (about ₹4,500) and Echo Dot (₹5,499 list) cost a third of the HomePod mini's ₹15,900. **Loyalty.** Airline and hotel programmes (Air India's Maharaja Club, Marriott Bonvoy), bank card points and retail programmes (Tata Neu, Reliance's One, Shoppers Stop First Citizen) all compete for the same wallet. Tata Digital, which runs Neu, BigBasket, Croma's online store, Tata 1mg and Tata CLiQ, grew revenue 12% in FY26 but its loss widened (Figure 9.9). **Public services.** Aadhaar covers almost every adult; UPI carries about 2,000 crore payments a month; DigiLocker has 67.6 crore users. Turnout in the Lok Sabha election rose sharply in 2014 and has since stayed in the mid-60s (Figure 9.8); young voters are the group that most often doesn't turn up.

Lok Sabha turnout
share of registered voters who voted

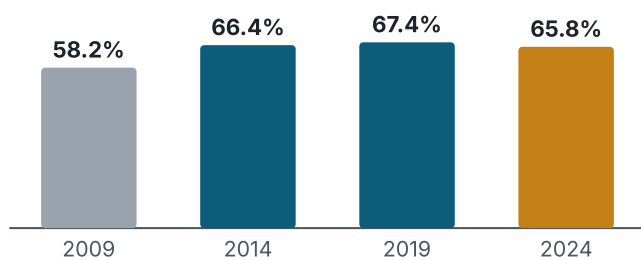


Figure 9.8 · Turnout has stalled. After the 2014 jump, turnout stayed flat and dipped in 2024. *Election Commission of India.*

Tata Digital: sales up, losses up
BigBasket, Neu, Croma online, 1mg, CLiQ

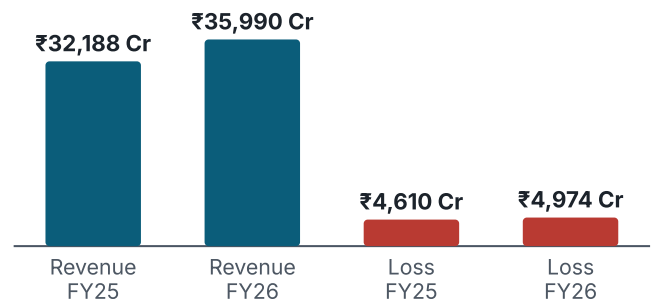


Figure 9.9 · More sales, bigger loss. BigBasket made up 64% of the FY26 loss. *Inc42, Business Standard (Sep 2026), from company filings.*

CHECK · "POINTS MAKE CUSTOMERS LOYAL" **OVERSTATED**

Most members of a loyalty programme are people who already shop there often, so a programme full of members proves little. Indians also carry several programmes at once, one per airline, bank card and store, so points rarely lock anyone in. Points change behaviour most near a reward. In a well-known study of a café's stamp card (Kivetz, Urminsky and Zheng, 2006), customers bought coffee faster the closer they got to the free one, and a card that started with two "free" stamps out of twelve was completed faster than a card needing ten out of ten. The lesson for design: make the next reward feel close and easy to reach, and measure the extra buying against members who didn't get the offer.

Turning points

Four changes shaped these businesses (Figure 9.10). **The iPhone and the App Store (2007–08)** turned a phone into a platform that earns after the sale. **Aadhaar (2010) and UPI (2016)** gave India shared rails for identity and money, and DigiLocker (2015) did the same for documents. **The Echo (2014)** showed a company could sell hardware near cost to plant a service, and its stall showed the risk when the service isn't used. **Tata Neu (2022)** tried to join one of India's largest groups under one app and one currency; by 2025–26 it had narrowed its focus to the co-branded card, payments, loans and loyalty.

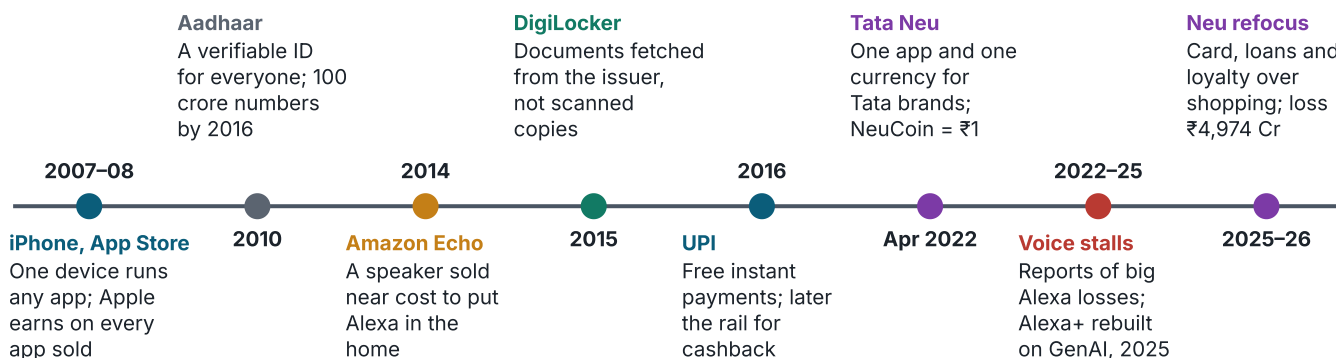


Figure 9.10 · Eight dates that changed these businesses. The winners built something people come back to; the stumbles sold a first transaction and hoped the habit would follow. *Company and government announcements; Tata Digital figures from Sep 2026 filings.*

The non-obvious mechanics

A device can't be patched like an app. Once a speaker ships, its microphones, chip and memory are fixed for years; only the software can change. So hardware PMs decide early which features need hardware (a better microphone for far-field voice) and which can arrive later by update. **Points are money the company owes.** A big balance of unspent points is a debt, and when a programme closes or changes, customers notice at once; in any merger of programmes, the old balances must move over at a fair rate. **Generous earn, narrow burn is a trap.** Points that can only be spent on a few items feel worthless, so breakage rises and the programme stops changing behaviour. **A public app competes with Instagram for attention,** but it has no ad budget and no reason to be opened most weeks, so it must be useful at the one moment that matters (the week before voting, the day of the traffic check). **Neutrality matters more than engagement.** A government app that ranks candidates or nudges one group more than another will be accused of taking sides, even when it doesn't.

IN THE INTERVIEW · QUESTIONS THIS SECTOR LOVES

- **“Price our new device.”** Ask who it is for and what the goal is (share or profit), find the cost floor, then price against what buyers compare it with, and count the services it brings.
- **“Design a loyalty programme.”** Work out what a point costs and who pays for it when it crosses brands, then make the points easy to spend; old balances must move over at a fair rate.
- **“Design an app to raise turnout (or tax filing, or vaccination).”** Find why people don't act, remove the friction at the moment of action, and measure the lift against a group that got no nudge.
- **“Why did this super-app struggle?”** Ask what brings people back each week; many brands under one login don't give anyone a reason to open it.

CASE 9.1 · AN APP TO RAISE VOTER TURNOUT

Product design Medium IIM A Product Mindset 2026–27, pp. 245–246

Try it first. Think of the last time you meant to do something civic (renew a licence, update an address, vote) and didn't. Was it that you couldn't, didn't care, or forgot? Now structure "design an app that gets more young people to vote" out loud, for five minutes.

THE QUESTION

Perugia's voting turnouts have stagnated at 55% in the last few years. The Government of Perugia has hired you to design an application to improve voter turnout. What are some of the features you would recommend for such an application?

A Okay. Can I start with a few clarifying questions about the problem?

Q Sure, go ahead.

A Why has turnout stagnated? Are the people skipping from a specific demographic? What are the current voting methods, and is the app meant to be a voting channel itself or only an engagement tool?

Q Those who skip cite a lack of knowledge of the candidates, low interest in the process, confusion about registration and voting, a sense that their vote doesn't count for much, and being busy on voting day.

Q The government's awareness campaigns reach only a few. The people skipping are mainly new voters aged 18–30. Voting remains a paper-ballot system and the government doesn't intend to change that. The app should attract more voters and engage them through the election season so they turn out.

A Since new voters are the most affected, I'll focus the app on them. The reasons cluster into three problems I can design for: a lack of information across the process, low interest, and the belief that one's vote doesn't matter. The third is important to call out separately, because information and engagement features alone won't shift it.

The remaining reason – being busy on voting day – I can only mitigate, not solve: voting stays a paper ballot (out of scope), so the app handles it through reminders and helping voters plan their booth visit rather than enabling remote voting. May I take a moment to think through the solution?

Q Sure, go ahead.

A Alright. Keeping the needs in mind, I have come up with the following features in the app: Addressing lack of information -

- **Candidate Scorecard:** This would be curated using the information from their previous work, their previous election performance, their headlines in the news, criminal and other charges against them. Since we are working with the government, some of this information can be directly sourced from the candidates, and the rest can be gathered from external sources.
- **App-based registration:** By digitizing the voter registration through the app, the voter would be able to register from anywhere using the app, reducing the time and money taken to set up and use an extensive offline registration system.
- **News section:** Capture the headlines to inform the user of the current issues in the news, which would be localized to their voting region.
- **Timeline of the voting process:** What is my booth and navigation to the voting booth.

Addressing lack of interest:

- **Gamification of the voters:** Running an app-based daily/weekly quiz to find the most informed voters. There may be perks associated with that, like preferential voting etc., depending on the rules and laws of the land.
- **Notification center:** Using notifications specifically to keep the voter engaged about the happenings of the process and the critical dates.
- **Connect with the candidates:** The candidates can leverage the app to communicate directly with their voters specific to the region. On the other side, the voters could ask questions and upvote the questions that they find relevant. It would be in the candidates' interest to answer more of the questions asked since it would only increase their reach, and the voters would be more engaged.
- **Non-transferrable badges for voters:** Perks like badges, QR to your exclusive photo at the booth etc., could be incorporated into the app to encourage voting. Do these look fine, or should I explore more features?

Q These should do. Could you elaborate on one of the user workflows that you see the user having on the application?

A A week before the election, when the app holds the most information: the user opens the app to their profile and gamified score (a percentile of how informed they are), the next quiz, and local news. They take the quiz, then open the candidate tab and select a re-contesting incumbent. The app shows the scorecard, recent work in office, latest headlines, and any legal charges. The user opens the "work" tab, which details recent parliamentary participation and bills presented, reads it, and closes the app – a flow where the user engages and consumes information.

Q Yes, that is good enough. Can you quickly provide some metrics that you would track for this application?

A The north-star metric is turnout lift – turnout among app users versus non-users, and overall turnout against the 55% baseline since that's what the government actually hired for.

Around it I'd track engagement (DAU/WAU, session duration, notification open rate), awareness (candidate-profile views per user, articles read, quiz completion), and conversion (voters registered through the app).

Q Thank you, that will be all.

WHAT WORKED · WHAT WAS MISSING

- ✓ **Asked why turnout stalled and who is skipping**, and got the five reasons and the segment (18–30, new voters) in one exchange.
- ✓ **Asked whether the app is a voting channel**, which fixed a big edge: paper ballots stay, so the app can only help people get to the booth.
- ✓ **Called out “my vote doesn’t count” as its own problem**, and said information features alone won’t shift it.
- ✓ **Tied the North Star to what the government hired for**: turnout against the 55% baseline.
- ✗ **“App users vs non-users” measures who installs, not what the app does**. Keen voters install voter apps. A random group that gets no reminders is the fair comparison.
- ✗ **Most features give information**; the research says the biggest effects come from reminders, plan-making and social norms, at the moment of action.
- ✗ **“My vote doesn’t count” was named, then left without a feature**.
- ✗ **Risky features**: a government-curated “scorecard” invites charges of bias; “preferential voting” as a quiz prize would break the rule of one person, one equal vote; app registration needs identity checks and a legal basis.
- ✗ **Never asked what exists today**. In India the Election Commission already runs the Voter Helpline app (name search, booth details) and campaigns to register young voters.

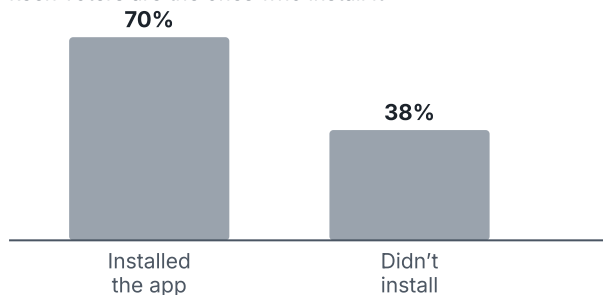
Scoping kit check: Confirm the thing ✓ **engagement tool, not a voting channel** · Goal ✓ **turnout** · Who ✓ **new voters, 18–30** · Today and the trigger ✗ **what the government already offers** · Edges ✗ **neutrality, privacy, what the law allows**. Design add-ons: journey stage ✗ · success metric at the end ✓. The costliest skip: edges. A government app lives or dies on looking neutral, and two of the eight features would fail that test.

CHECK · “THE NORTH STAR IS TURNOUT AMONG APP USERS VERSUS NON-USERS” DOESN'T HOLD UP

Picture two groups on polling day. The people who downloaded a voter app were already interested: many would have voted without it. Say 70% of app users vote and 38% of everyone else. The 32-point gap mostly shows who installs the app (Figure 9.11). The fair test is a **holdout**: pick a random 10% of app users and send them no reminders. Both groups are the same kind of people, so the gap between them (say 43% vs 40%) is what the reminders caused. Overall turnout against 55% is still the right headline, but it moves for many reasons at once: the candidates, the weather, the mood of the election.

Misleading: app users vs everyone else

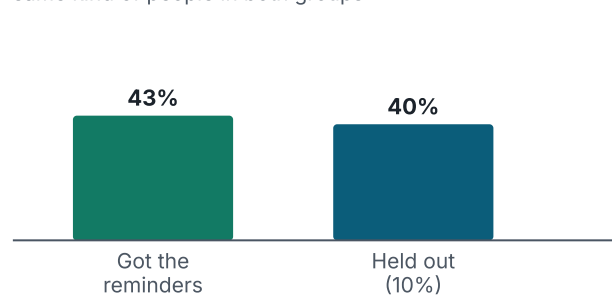
keen voters are the ones who install it



Gap of 32 points, but most of it is who installs, not what the app did

Fair: a random 10% get no reminders

same kind of people in both groups



Gap of 3 points: this is what the reminders actually caused

Figure 9.11 · Two ways to measure the app, one of them fair. Comparing users with non-users credits the app for people who would have voted anyway. A random holdout keeps the groups alike, so the gap is the app's real effect. *Illustrative numbers.*

STRONGER ANSWER

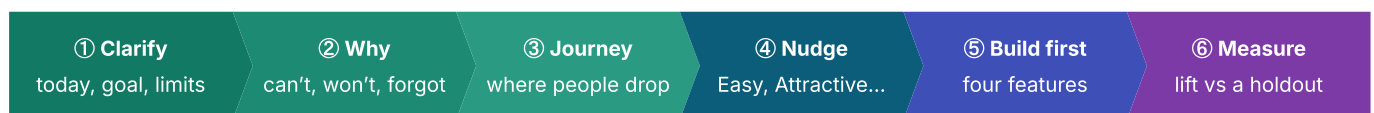


Figure 9.12 · Six moves for “design an app to change a behaviour”. Find out whether people can't, won't or forget, fix the step where most drop out, and measure against a group that got nothing. *Author's summary.*

A ① CLARIFY

Three quick questions. What does the government offer today: can people check their name and booth online? How many 18–30-year-olds are on the roll, and how many voted last time? And what are the limits: the app must be neutral between parties, and I assume we can't share anyone's data with candidates.

Q There's a website to search your name; nothing on mobile. About 30,000 of the city's 1.5 lakh voters are 18–30, and about 40% of them voted, against 59% of older voters. Yes, strictly neutral, and no data sharing.

Why this move: what exists today tells you what to build on; the size of the group tells you how much a feature can move the city's number; the limits rule out features before you fall in love with them.

A ② WHY

I'd sort the five reasons into three kinds. Some people **can't**: they aren't registered or don't know their booth. Some **won't**: no interest, or they think their vote won't count. Some **forget** or get busy. They need different fixes. If a survey shows reasons like these (Figure 9.13), "busy or forgot" and "unsure how" together are over half, and those are the easiest for an app to fix.

Why this move: a person who can't vote needs a step removed, one who forgets needs a prompt, and one who doesn't care needs a reason. Lumping them together produces a feature list that fits nobody.

Why young people on the roll didn't vote last time: share of reasons given (illustrative survey)

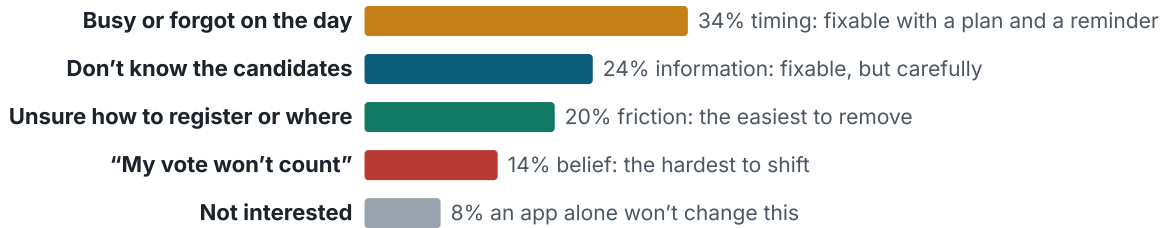


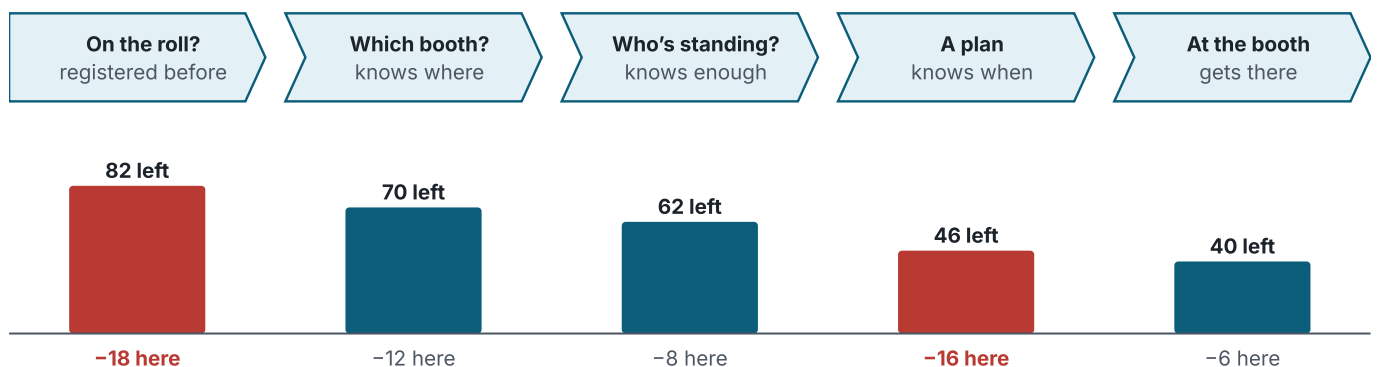
Figure 9.13 · Why young voters stayed home. The reasons sort into can't (register, booth), won't (interest, belief) and forgot (timing). The orange and green bars are the ones an app can fix cheaply. *Illustrative split of the five reasons named by the interviewer.*

A ③ JOURNEY

Next I'd follow one young voter from registration to the ballot box and ask where people drop out (Figure 9.14). Two leaks stand out. Many miss the registration deadline. And many mean to vote but never decide when, so the day fills up with other things. Learning about candidates matters too, but it loses fewer people.

Why this move: putting the reasons on a journey shows where to act. It also shows that the candidate scorecard, the most elaborate feature on the candidate's list, sits at a step that loses the fewest people.

100 young people on the roll: how many are still on track after each step (illustrative)



Red bars mark the two biggest leaks: registering in time, and turning a vague intention into a plan.

Figure 9.14 · The voter's journey and where people drop out. Of 100 young people on the roll, 40 vote. The biggest leaks are registering in time and making a plan for the day. *Illustrative numbers, chosen to match 40% turnout.*

A ④ NUDGE

For each leak, the smallest nudge that works. **Registration:** a reminder four weeks before the deadline with a direct link to the official form and a checklist of documents. **Booth:** one tap to see "you're on the list, booth 14, Government School, open 7 am to 6 pm", with the route. **Plan:** three days before, the app asks "What time will you go, and from where?" and saves the answer to the phone's calendar. **Polling day:** a morning message: "Your booth opens at 7. You said 9 am." For candidates, the same facts for everyone, taken from their official declarations (education, assets, pending cases), in the same format, with no scores.

Why this move: each feature sits on a leak and is small enough to build before the election. Using the candidates' own sworn declarations keeps the government out of judging them.

A ⑤ BUILD FIRST

With one election as the deadline, I'd ship four things: the list and booth check, registration reminders, the plan-making prompt with the calendar entry, and the polling-day message. Candidate facts come next, because they need careful checking. I'd drop quizzes with prizes, a news feed and candidate Q&A: each needs moderation, and each invites a charge of bias. For "my vote won't count", one honest screen: "In the last city election, ward 7 was decided by 212 votes."

Why this move: a public app has one deadline it can't move. Choosing four features that remove friction at the moment of action gives the best chance of an effect this election.

A ⑥ MEASURE

Goal (North Star): turnout of 18–30s who got the reminders, against a random group who didn't (the holdout), checked on the official record of who voted. **Sizing:** if the app and SMS reach 18,000 young voters and lift their turnout by 3 points, that's $18,000 \times 3\% = 540$ extra votes, about 0.4 points on the city's 55%. To see a 3-point lift from 40%, each group needs about $16 \times 0.4 \times 0.6 \div 0.03^2 \approx 4,300$ people, so the holdout of 1,800 is too small; I'd hold out 4,300 in year one. **Safety checks (guardrails):** complaints of bias, wrong booth details, and opt-outs from messages. All numbers illustrative.

Why this move: an honest size stops the government expecting 55% to become 70%. The sample-size check shows the test can tell a real effect from noise, and the guardrails protect trust, which is the app's only asset.

HOW THE ANSWER HANGS TOGETHER

The clarifying questions gave the base (30,000 young voters at 40%) and the limits (neutral, no data sharing). Sorting reasons into can't, won't and forgot pointed at the two big leaks: registering in time and making a plan. Each feature sits on one leak and is small enough to build before polling day. The North Star is turnout against a random holdout, because that is the only number that shows what the app caused; bias complaints and wrong booth details are the guardrails, because a public app runs on trust.

FRAMEWORK · NUDGES, USING EAST (EASY, ATTRACTIVE, SOCIAL, TIMELY)

1 · The situation. Most young people in Perugia intend to vote; 60% of them don't. Information alone rarely closes that gap between meaning to and doing. What closes it is making the action easier, more appealing, visible among peers, and asked for at the right moment. That is what a **nudge** does: it changes how a choice is presented without banning anything or paying anyone.

2 · An everyday example. A family that keeps forgetting their vitamins. Telling them vitamins are healthy doesn't help; they know. What helps: the bottle next to the toothbrush, a sticker chart for the children, "everyone at school takes them", and a reminder at breakfast.

	Vitamins at home	Voting in Perugia
Easy	Bottle next to the toothbrush	One tap to see your booth, hours and route
Attractive	A sticker chart the children like	A first-time-voter pledge card; the inked-finger photo frame
Social	"Everyone at school takes them"	"8 in 10 first-time voters in your ward checked their booth"
Timely	A reminder at breakfast, not at night	Plan three days before; a message on polling morning

3 · The framework. EAST was set out by the UK's Behavioural Insights Team in 2014 as a checklist for designing nudges (Figure 9.15). **Easy:** remove steps, fill things in by default, make the next action one tap. **Attractive:** draw attention and make the action feel rewarding. **Social:** show that people like me do it, or let people commit in front of friends. **Timely:** prompt when people are most likely to act, and help them plan. The evidence from voting is useful because the effects were measured against control groups (Figure 9.16): telling people that neighbours would see who voted raised turnout by about 8 points; a phone call asking when and how someone would vote raised it by about 4; a plain "voting is your duty" letter by under 2.

Using it, step by step (on this case):

1. Name the exact action and the moment: "vote at booth 14 on polling day".
2. Map the journey to that moment and find the biggest leaks: registering in time, making a plan.
3. For each leak, ask the four questions in order: can it be easier, more appealing, more social, better timed?
4. Prefer the smallest change that removes a step (booth in one tap) over the cleverest message.
5. Test each nudge against a random group that doesn't get it, and keep only what moves the number.

COMMON SLIP · nudging people who can't act. A reminder to vote is useless to someone who isn't on the roll; fix the "can't" before the "forgot".

E	Easy Remove steps	One tap: "Am I on the list? Which booth?", with the route and the booth's opening hours
A	Attractive Make it appealing	A first-time-voter pledge card to share; the inked-finger photo frame after voting
S	Social Show what others do	"8 in 10 first-time voters in your ward have checked their booth" (only if true)
T	Timely Ask at the right moment	Three days before: "What time will you go?" Polling morning: "Your booth opens at 7"

Figure 9.15 · EAST applied to voting. Each letter is a question to ask about one step of the journey. The examples are small on purpose: the effect comes from the moment and the ease, not from the size of the feature. *EAST from the Behavioural Insights Team (2014); examples are the author's.*

Extra turnout from famous experiments, in percentage points over a group that got nothing

"Your neighbours will see"	+8.1 points	mailer listing who voted (US, 2006)
"Here is your voting record"	+4.9 points	mailer showing your own past turnout
A call to make a plan	+4.1 points	"when, from where, how?" (US, 2008)
"Voting is your duty"	+1.8 points	a plain civic-duty letter
Friends who voted, on Facebook	+0.4 points	tiny per person, but 6.1 crore people

Figure 9.16 · What moved turnout in real experiments. Social pressure worked best but can feel like shaming; plan-making worked well with no downside; the Facebook effect was tiny per person but reached 6.1 crore people. *Gerber, Green and Larimer (2008); Nickerson and Rogers (2010); Bond et al. (2012). US elections; effects in India will differ.*

- 4 · When it misleads.** A social message can backfire when the norm is low. “Only 40% of young people vote” tells a young reader that not voting is normal. Pressure can also turn into shaming: publishing who didn’t vote would raise turnout and anger a city. And nudges are small. A PM who promises that an app will lift turnout from 55% to 70% will be judged against that promise; a lift of 1 to 4 points among the people reached is a good result.
- 5 · In the interview.** “Most of them mean to vote but don’t, so I’d fix the two steps where they drop out, registering and planning the day, with a one-tap booth check and a plan-making prompt, and I’d measure the lift against a random group that gets no reminders.”

CASE 9.2 · PRICING AN APPLE SMART SPEAKER

Pricing Medium IIM B Sigma PM Casebook 2026–27, p. 93

Try it first. A friend with an iPhone asks whether to buy an Echo Dot at ₹5,499 or an Apple speaker. What would the Apple speaker have to do, and cost, for her to choose it? Now price “Apple Home” out loud, for five minutes.

THE QUESTION

How would you price Apple Home, the smart speaker from Apple?

- A** Just to be clear, Apple Home is a smart speaker which in addition to its music playing capabilities, has an AI assistant which can take voice commands and fetch data over the internet. Am I correct with this?
- Q** Yes, you are correct. One thing I would like to point out – Apple Home can also connect with and control the smart home devices in the house..
- A** Got it. From what I understand, these capabilities make it a competitor to smart speakers such as Alexa and Google Home. Is Apple Home a new launch by Apple or is it already present in the market?
- Q** We are looking to launch Apple Home in the next quarter. We hope to wrestle away at least 20-30% market share from other players in the market
- A** Got it! Since this is a new product launch, one strategy could be to price Apple Home lower than our competitors to capture market share. However, historically, Apple’s strategy in pricing has been to maintain high differentiation in their products in order to ask for premium prices. Consumers associate Apple’s brand with high-premium prices because they are used to Apple’s innovative product designs and quality. Therefore, pricing a future Apple Home device should follow this same high-differentiation-premium price strategy to preserve the brand’s leverage. To come up with the numbers, can you help with the products and prices our competitors Google and Amazon offer?
- Q** Sure, let me give you the prices of our competitor products. Google Nest is priced at Rs 7K while Google Nest Mini is priced at Rs 5K. For Alexa, we again have multiple versions ranging from at Rs 5.5K to Rs 10K.
- A** Got it. First of all, just like our competitors, I would introduce 2 variants of Apple Home in the market. Offering a lower priced device would allow us to hit competitors both in the low end and high end of the market. This strategy is especially useful in India where most of the users are price sensitive. This would entice customers who want a premium Apple product at a lower price.
- Secondly, I want to calculate the premium Apple charges for its products to arrive at the exact price point. Just to have an estimate of the kind of premium Apple charges, let’s compare the Google Pixel 8 and Apple iPhone 15. Now they have comparable specs but from the prices I remember the iPhone price is 30% more. So, we can take around 30% as the premium for Apple products over similar Google products. Using this we come at the price level of Rs 6.5K for Apple Home Mini and Rs 13K for Apple Home.
- Q** I love the fact that you brought in so many numbers into the analysis. You mentioned that Apple justifies its higher price through product differentiation. What are the features you would offer as differentiation?
- A** First let’s talk about the key features of Alexa and Google Home. According to me, Alexa’s key value proposition is its integration to Amazon’s online store. This makes online shopping with Alexa seamless. For Google Home, the integration with Google’s search engine is the key highlight. I would not want to compete with these devices on their strong points.
- 1) For Apple iPhone, the voice assistant Siri has been a huge success and hence becomes an obvious place where we can innovate. I would enhance personalisation in Apple Home. Usually, the device is used by multiple people in the house. Apple Home should store data mapped to different users and give personalised replies. These different users should be identified by their voice. For example, let’s take two users in a house who have had a historical interest in sports and politics respectively. When they get up in the morning and ask Apple Home to “Tell me the latest news”, they should be given different answers.
- 2) Another of Apple’s strong points has been its innovative design. There are numerous stories of Steve Jobs not going ahead with the design of iPhone and iPod until it was absolutely perfect. We should do something similar in Apple Home as well.
- Q** That was a thorough analysis. I really liked your approach. Thank you!

WHAT WORKED · WHAT WAS MISSING

- ✓ **Confirmed the product and asked new or existing**, and learnt it controls smart-home devices too.
- ✓ **Saw the tension** between a low launch price for share and Apple’s habit of premium prices.
- ✗ **Didn’t challenge the goal.** 20–30% share and a premium price pull in opposite directions; the candidate should have asked which one wins.

- ✓ **Asked for competitor prices** and proposed two variants, as rivals do.
- ✓ **Refused to fight Alexa and Google on their strengths** (shopping, search).

- ✗ **Missed who can buy it.** An Apple speaker is set up with an iPhone, and iPhones are under a tenth of phones in use in India (industry estimates, 2025–26). The market is iPhone homes, not all homes.
- ✗ **The 30% premium is wrong twice:** the phone comparison is off, and Apple's actual speakers cost far more than 1.3 times the rivals.
- ✗ **No cost floor:** ₹6,500 is less than what the speaker costs Apple once GST and the shop's margin are paid.
- ✗ **The differentiator already exists.** HomePod has recognised different voices in a home since 2019; Alexa (Voice ID) and Google (Voice Match) do it too.

Scoping kit check: Confirm the thing ✓ · Goal ✗ **took “20–30% share” at face value** · Who ✗ **iPhone owners only** · Today and the trigger ✓ **new launch** · Edges ✗ **which market**. Pricing add-ons: competitor prices ✓ · cost floor ✗ · plan structure ✓ **two variants** · share or profit ✗. The costliest skip: who. Once the buyers are iPhone homes, the share target and the price both change.

CHECK · “APPLE CHARGES ABOUT 30% MORE THAN GOOGLE FOR COMPARABLE PRODUCTS, SO PRICE 30% OVER RIVALS” DOESN'T HOLD UP

In India the iPhone 15 launched at ₹79,900 and the Pixel 8 at ₹75,999: about 5% apart, not 30%. More to the point, Apple's own speakers are nowhere near 1.3 times the rivals. On Apple's India store in September 2026 the HomePod mini is ₹15,900 and the HomePod ₹44,900, against about ₹4,500 for a Nest Mini and ₹5,499 for an Echo Dot (Figure 9.17). Apple doesn't add a fixed markup to a rival's price. It prices for people who already own an iPhone and value how its devices work together, and it accepts a small share of the market in return. A price copied from another product category tells you nothing about this one.

Smart speakers in India, by price (orange: the candidate's proposal)

Google Nest Mini	₹4,499 street price, 2026
Amazon Echo Dot	₹5,499 list price, 2026
Candidate's "Home Mini"	₹6,500 Nest × 1.3: under Apple's cost floor
Google Nest (casebook)	₹7,000 the interviewer's figure
Top Echo (casebook)	₹10,000 the interviewer's figure
Candidate's "Home"	₹13,000 Echo × 1.3
Apple HomePod mini	₹15,900 Apple India store, Sep 2026
Apple HomePod	₹44,900 Apple India store, Sep 2026

Figure 9.17 · Smart speakers in India, by price. The candidate's prices (orange) sit among the rivals; Apple's real speakers cost several times more, because Apple sells to a different buyer. *Apple India store (Sep 2026); Flipkart and retailer listings (2026); casebook figures as given by the interviewer.*

STRONGER ANSWER



Figure 9.18 · Six moves for “price our new device”. Settle the goal and the buyer first; then the price has a floor (cost) and a ceiling (what buyers compare it with). *Author's summary.*

A ① CLARIFY

Two things decide the price, so I'd ask them first. Which market: India, or everywhere? And the goal: 20–30% share and a premium price don't usually go together, so if we must choose, is it share or profit? One fact I'd check: does it need an iPhone to set up, like Apple's other speakers?

Q India first. Yes, it needs an iPhone. Leadership says share, but it must make money.

Why this move: “share or profit” changes the answer from ₹6,500 to ₹15,000, so it has to be settled before any number. The iPhone question sets the size of the market.

A ② WHO BUYS

Then the buyers are iPhone homes. Say 20 lakh smart speakers sell in India a year and one in ten goes to an iPhone home: about 2 lakh. Even winning a quarter of those is 50,000 speakers, about 2.5% of the whole market. So I'd restate the goal as **share of iPhone homes that buy a speaker**: 25–30% of them is ambitious and possible; 25% of all speakers isn't. All figures illustrative.

Why this move: a share target only makes sense against the people who can buy the product. Restating it keeps the goal and the price from fighting each other.

A ③ COST FLOOR

Next, the lowest price that doesn't lose money. Say parts, assembly, freight and warranty cost ₹6,300 for the mini. The shop keeps about 12% of the price before tax, and GST is 18% on top. So the break-even shelf price is $₹6,300 \div 0.88 \times 1.18 \approx ₹8,450$. At ₹6,500, Apple loses about ₹1,450 on every speaker. Illustrative costs.

Why this move: in India the price on the box includes tax and the shop's cut, so a device that "costs ₹6,300" can't sell at ₹6,500. Working back from cost stops a launch that loses money on each unit.

A ④ VALUE

An iPhone owner compares it with an Echo Dot at ₹5,499 and with doing nothing. What she gets that an Echo doesn't: it plays what her phone plays with a tap, sets up in seconds, keeps voice requests private, and runs Apple's smart-home hub. What it earns Apple after the sale: more Apple Music and iCloud subscriptions, and a household more tied to Apple. I'd count that in rupees: say 30% of buyers add a ₹100-a-month music plan, about ₹1,000 of profit a speaker over three years (Figure 9.19).

Why this move: the value is set by the buyer's other options, and the services only justify a lower price if they are counted in rupees and actually happen.

One mini speaker at ₹12,900, over three years (illustrative)

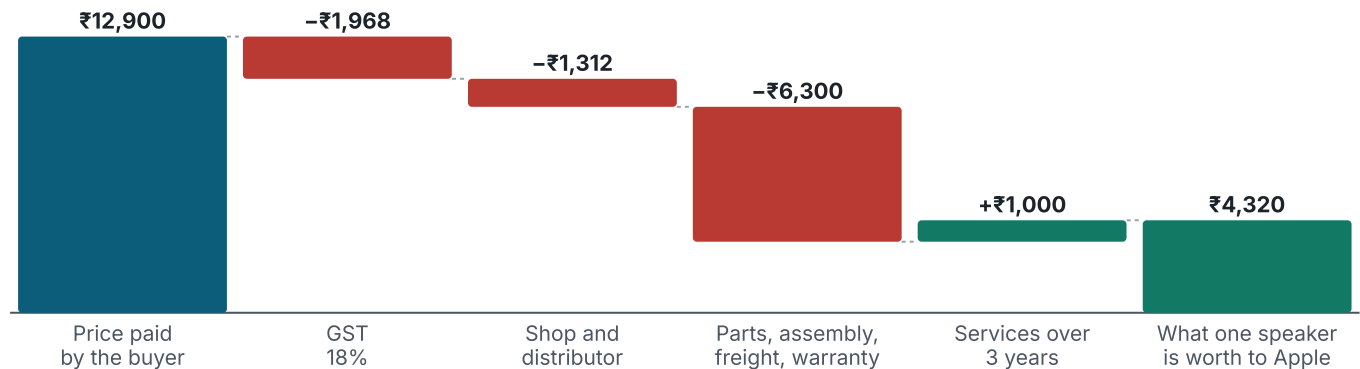


Figure 9.19 · What one speaker at ₹12,900 is worth to Apple. Tax and the shop take about a quarter; services over three years add a little on top. Services help, but they don't turn a loss-making price into a good one. *Illustrative costs and take-up.*

A ⑤ PRICE

Now compare profit at four prices (Figure 9.20). At ₹6,500 it loses money even with services. At ₹15,900 each speaker earns most, but fewer homes buy. At ₹12,900 total profit is about the same as at ₹15,900, and it reaches 55,000 homes instead of 35,000. So: **"Home mini" at ₹12,900 and "Home" at ₹29,900**, and a stereo pair of minis at ₹23,900. Launch with six months of Apple Music included, and sell at Apple Stores and Croma, where iPhone buyers already shop.

Why this move: the best price is where total profit peaks, and near the peak the curve is flat. When two prices earn about the same, pick the one that puts the device into more homes, because each home adds to the ecosystem.

Mini speaker in India: profit to Apple over 3 years at four prices (₹ crore, illustrative)

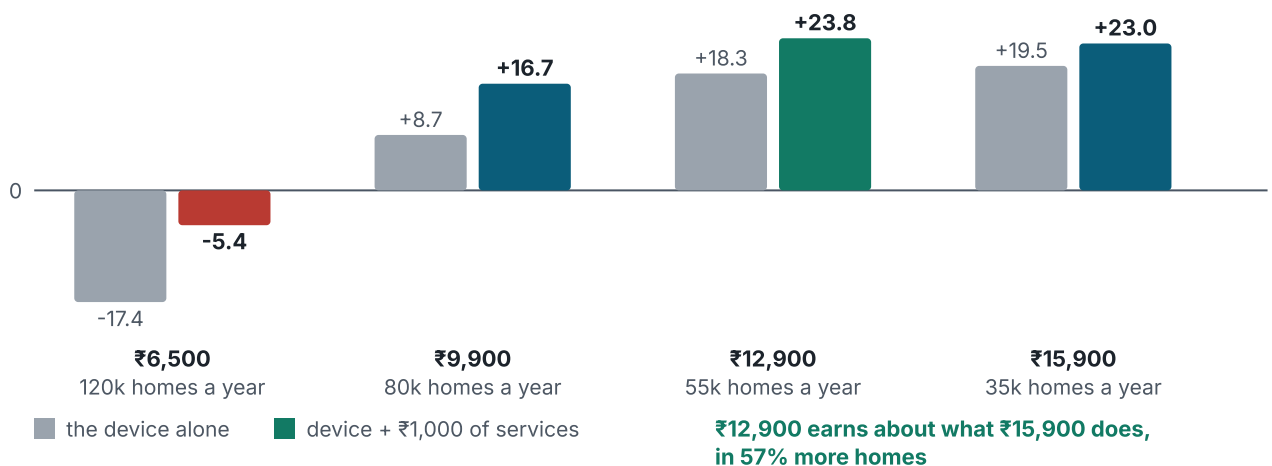


Figure 9.20 · Profit at four prices. The cheap price loses money; the two premium prices earn about the same in total. The lower of the two reaches more homes, which is worth more to Apple over time. *Illustrative demand, costs and services.*

Variant	For whom	What's different	Price (illustrative)
Home mini	A bedroom or a first speaker	Voice assistant, music from the iPhone, smart-home control	₹12,900
Two minis	A living room on a budget	Paired as left and right speakers	₹23,900 (save ₹1,900)
Home	A family room; the home's hub	Fuller sound, room sensing, the always-on smart-home hub	₹29,900

A 6 MEASURE

Goal (North Star): share of iPhone homes that bought a smart speaker this year and chose ours. **Inputs:** sales per week at each price; share of buyers who pair a second speaker; Apple Music trials that become paid. **Safety checks (guardrails):** gross margin per unit over 30% (₹12,900 gives about 34%); returns within 14 days under 4%; complaints that it doesn't understand Indian accents or Hinglish. **Trade-off:** at ₹12,900 we give up some margin per unit for more homes; I'd review after two quarters. All figures illustrative.

Why this move: the North Star follows the restated goal, so the team isn't judged on a share of homes it can't sell to. The accent guardrail matters in India: a speaker that mishears is returned, however good the price.

HOW THE ANSWER HANGS TOGETHER

Asking the goal and the setup requirement turned "20–30% of all speakers" into "a quarter of iPhone homes". The cost floor (₹8,450) ruled out the candidate's ₹6,500. The rivals and the services set the value; comparing profit at four prices showed ₹12,900 earns about what ₹15,900 does in 57% more homes. The North Star counts the homes Apple can actually reach, and the margin and returns guardrails keep the lower price honest.

KEY IDEA · PRICE FOR THE BUYER WHO CAN BUY, THEN COUNT THE WHOLE LIFE

A premium device is priced against what its own buyers compare it with, not against the whole market. Then look at the whole life of one unit: price, less tax and the shop's cut, less cost, plus what it earns afterwards. The services are real only when they are counted in rupees; "it builds the ecosystem" on its own is not a reason to sell at a loss.

CASE 9.3 · ONE LOYALTY PROGRAMME FOR THE TATA GROUP

Go-to-market

Medium

IIM A Product Mindset 2026–27, pp. 285–286

Try it first. Count the Tata brands you or your family paid last year: groceries, a watch, a flight, a coffee, a phone. Would one set of points across all of them change where you shop? Who should pay for the points? Structure the programme out loud, for five minutes.

THE QUESTION

The tata group wants to incentivize their customers and want to increase their loyalty towards the brand. We need your help to prepare a loyalty program for the conglomerate.

A To make sure I understand the problem correctly, we're designing a single loyalty program across all Tata brands-such as Taj, Titan, Tanishq and automobiles.

Q Yes, that is correct.

A Many Tata brands already have their own loyalty programs. What happens to those?

Q You can assume that all the existing loyalty programs will be replaced by the new loyalty program that you would suggest.

A One concern: customers have already earned reward points. Invalidating them would create a poor customer experience.

Q Good that you mentioned it. In your solution, ensure that there is a smooth transition for the existing reward system.

A Sure. One last clarification before I begin-will brands have flexibility in rewarding customers, or will Tata Group control everything centrally?

Q We would like you to think how this Loyalty program should be structured. Also, comment on its technical feasibility.

A Sure. May I take a couple of minutes to structure my thoughts?

Q Sure.

A I'd divide the solution into three parts:

- Reward structure
- Technical implementation
- Customer experience

For the reward structure, I think there should be three levels:

- Tata Group Points - Common rewards across all Tata brands.
- Brand Points - Individual brands can run their own reward programs.
- Store Points - Local stores can run city/store-specific promotions.

A customer can earn one or more of these depending on the campaign.

Q Makes sense, carry on.

A This hierarchy gives flexibility while keeping the experience unified.

- Tata Group manages ecosystem-wide rewards.
- Brands control brand-specific campaigns.
- Stores can run local offers when needed.

Q Usually when the campaign runs, the points are provided in a particular city. How will you ensure that?

A Instead of asking every store to configure campaigns:

- Configure offers at the brand or Tata level.
- Use the Store ID to determine eligibility.
- Stores automatically inherit the applicable campaigns.

This keeps campaign management scalable.

Q That seems like a good plan. Can you explain the implementation plan?

A I'd use the customer's mobile number as the primary identifier because it's:

- Easy to remember
- Already available across brands
- Suitable for creating a unified customer profile

One limitation is that customer identity will exist at the Tata level instead of the brand level. To solve this, every brand should register customers against the common Tata ID.

Q Can you suggest other things to consider for a good loyalty program?

A The real value of loyalty comes from redemption. I'd enable:

- Redemption within the same brand.
- Cross-brand redemption wherever possible.

For example, points earned at Titan could also be redeemed at Starbucks.

Q Interesting, anything else you'd like to add?

A One last point, we discussed the existing reward points at the beginning.

I'd ensure a smooth migration by:

- Mapping all existing reward points to the new customer ID.
- Migrating historical customer data.
- Running a one-time data migration before launch.

This ensures customers don't lose their rewards while transitioning to the new system.

Q Thank you for the insights.

WHAT WORKED · WHAT WAS MISSING

✓ **Confirmed the scope** (one programme for all brands) and asked what happens to existing programmes.

✓ **Raised the old points early**, before the interviewer did, and closed with a migration plan.

✓ **A clear three-level structure** (group, brand, store) with campaigns set centrally and inherited by stores.

✓ **Saw that redemption is where the value is**, including across brands.

✗ **This programme already exists.** Tata Neu launched in April 2022 with NeuCoins (1 NeuCoin = ₹1) across Tata brands. The candidate never asked what exists or why it hasn't worked as hoped.

✗ **No economics.** What a point costs, how many are redeemed, how much must sales rise to pay for them.

✗ **"Points earned at Titan, redeemed at Starbucks" skips the hard part:** who pays Starbucks for the coffee.

✗ **The mobile number as ID** ignores recycled numbers, shared family phones and consent to share data across separate companies.

✗ **No goal or success metric;** "loyalty" was never turned into a number.

Scoping kit check: Confirm the thing ✓ **one programme, all brands** · Goal ✗ **never defined** · Who ✗ **which customers** · Today and the trigger ✗ **Tata Neu already exists** · Edges ✓ **old points, central vs brand control**. GTM add-ons: standalone or built on existing assets ✗ · competitors ✗ · budget ✗. The costliest skip: today. Designing from scratch a programme that has run since 2022 misses the only useful question: what has it taught us?

CHECK · "USE THE CUSTOMER'S MOBILE NUMBER AS THE PRIMARY IDENTIFIER" **DIRECTIONALLY RIGHT**

The number is the one thing every Tata brand already collects, so it is the right key for matching records. Three things need handling. **Recycled numbers:** operators reissue disconnected numbers after a few months, so a new owner could see someone else's points; log-in needs an OTP and a check on recent activity before a big redemption. **Shared phones:** in many homes one number is used by the whole family; allow family members under one account. **Consent:** Titan, Tata Consumer and Air India are separate companies, so under the Digital Personal Data Protection Act, 2023 a customer must agree, in plain words, before one brand's purchase history is shared with another.

STRONGER ANSWER

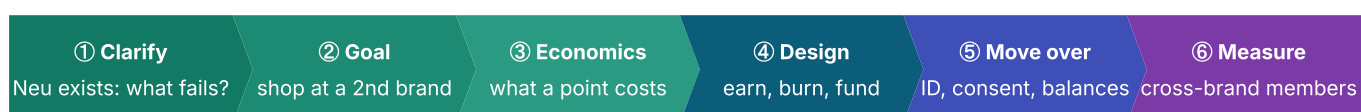


Figure 9.21 · Six moves for "design a loyalty programme". Start from what exists, turn loyalty into one behaviour, and price the points before designing them. *Author's summary.*

A ① CLARIFY

Tata Neu and NeuCoins have run since 2022. Is this a redesign of Neu, or something new? And what has Neu taught us: do members earn but not spend, or shop at one brand only?

Q

Treat it as the redesign. Members join, but most shop at one brand only, and many have coins they never spend.

Why this move: a programme with years of data is a gift. "Most shop at one brand" tells you the goal; "coins never spent" tells you the points don't feel valuable.

A ② GOAL

Then I'd define loyalty as one behaviour: **a member buying from a second Tata brand**. A group programme only earns its cost if it moves money between brands; a Tanishq customer who stays loyal to Tanishq could be rewarded by Tanishq alone. Target the people most likely to cross: BigBasket and Croma shoppers who already buy often, shown the brand that fits their life (a coffee near the office, a hotel for a long weekend).

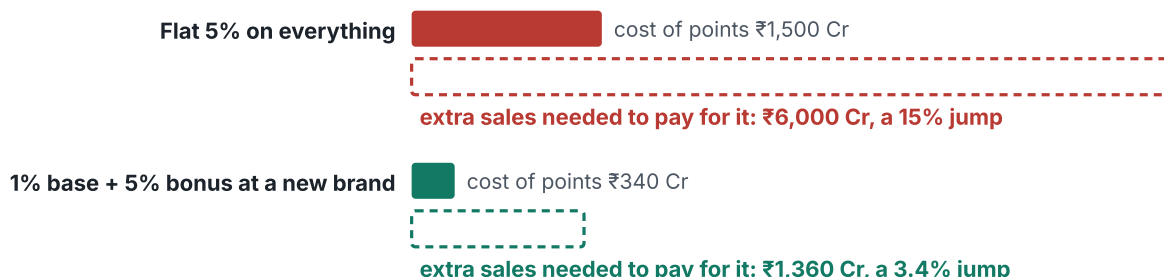
Why this move: "increase loyalty" can't be measured or designed for. A single behaviour tells you whom to reward and what to count.

A ③ ECONOMICS

Say 1 crore active members spend ₹40,000 a year each: ₹40,000 crore of sales. A flat 5% everywhere, with 75% redeemed, costs $40,000 \times 5\% \times 75\% = \text{₹1,500 crore a year}$. At a 25% gross margin, sales must rise by ₹6,000 crore, 15%, just to break even (Figure 9.22). Most of those points go to people who would have bought anyway. A 1% base plus a 5% bonus on the first ₹10,000 at a new brand, for 10 lakh members who try one, costs about ₹340 crore and needs a 3.4% rise. All figures illustrative.

Why this move: a point is a discount. Working out what it costs, and how much extra buying it needs, shows that a flat generous rate mostly pays people for what they would have done anyway.

One year, 1 crore members spending ₹40,000 each = ₹40,000 crore of sales (illustrative)



Cost = sales × earn rate × 75% redeemed. Extra sales needed to pay for it = cost ÷ 25% gross margin.

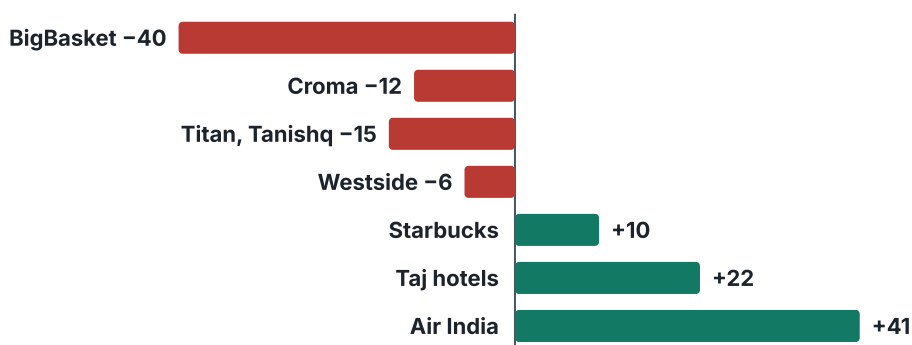
Figure 9.22 · A generous flat rate vs paying for the behaviour you want. A flat 5% needs sales to jump 15% to pay for itself. Paying mainly for a first purchase at a new brand costs a fifth as much. *Illustrative members, spend, margin and redemption.*

A ④ DESIGN

Earn: each brand sets its base rate from its own margin: 0.5–1% on groceries, 2–3% on fashion and electronics, more on hotels and flights. **Burn:** coins spendable at every brand, in any amount, at checkout, so a coin always feels like ₹1. **Fund:** the brand that issues a coin pays for it; the brand that accepts it is paid from a group pool, less a small fee; the group funds the new-brand bonus from marketing; and the co-branded HDFC card buys coins for card spends. **Watch the flows** (Figure 9.23): coins earned on weekly groceries get spent on flights and hotels, so BigBasket pays into the pool every month. If its earn rate is set too high, it will quit.

Why this move: a group programme holds together only if every brand sees its own maths work. Earn rates tied to each brand's margin and a clear settlement rule keep the brands in.

Net settlement per brand in a year: pays into the pool (red) or is paid from it (green), ₹ crore (illustrative)



Brands where points are earned pay the brands where they are spent. Set each brand's earn rate from its own margin, or the grocery business ends up paying for other people's holidays and quits the programme.

Figure 9.23 · Where the money moves between brands. Frequent, low-margin purchases earn most coins; rare, high-value ones absorb them. The pool must balance to zero, and each brand must accept its side. *Illustrative amounts.*

A ⑤ MOVE OVER

ID: the mobile number, with an OTP, a family account for shared phones, and a check before large redemptions to catch recycled numbers. **Consent:** one plain screen at sign-up: “Let Tata brands use your purchases to send you offers?”, which a member can turn off any time. **Old balances:** convert at their rupee value, not one-for-one: 1,000 old points worth ₹250 become 250 coins. Tell members 90 days ahead, show the conversion in the app, and don’t expire converted coins for a year.

Why this move: moving money people already own is where trust is won or lost. Converting at value is fair to members and to the brand paying for the points; a year’s grace prevents a wave of complaints.

A ⑥ MEASURE

Goal (North Star): share of active members who bought from two or more brands in the last 90 days; say from 12% to 20% in a year. **Inputs:** coins redeemed as a share of coins earned; members who try a new brand with the bonus; repeat purchase at that brand within 90 days. **Proof it pays:** a random 5% of members who get no bonus, to measure the extra spend the bonus caused. **Safety checks (guardrails):** total cost of coins under 1% of group sales; consent opt-out rate; complaints about missing coins. Illustrative targets.

Why this move: the North Star counts the behaviour the programme exists for. The holdout proves the bonus caused extra buying, and the cost cap keeps the brands willing to fund it.

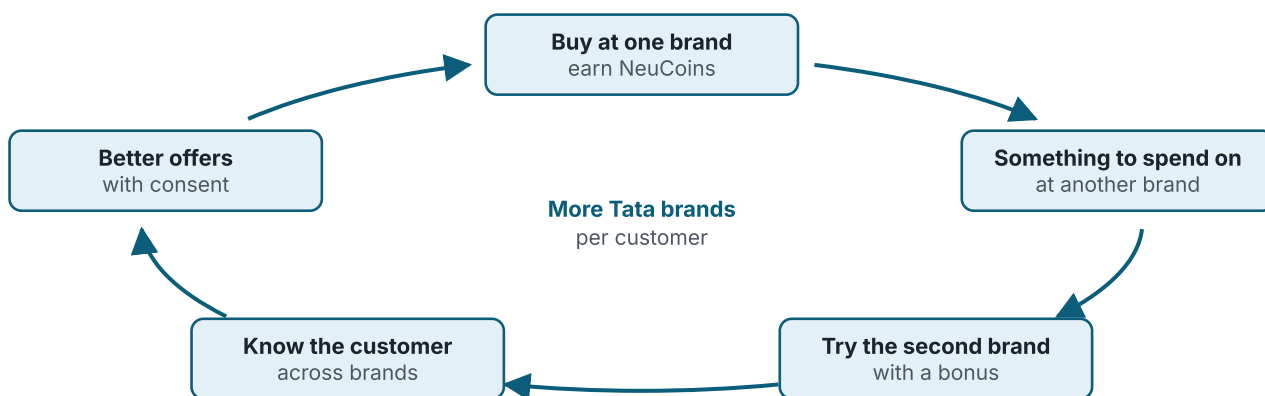


Figure 9.24 · The loop a group programme needs. Each turn adds a brand to a customer’s life; the loop breaks wherever coins feel worthless or there is nothing tempting to spend them on. *Author’s diagram.*

HOW THE ANSWER HANGS TOGETHER

Asking what exists turned a blank-slate design into a redesign of Tata Neu with a known problem: members stick to one brand and don’t spend their coins. That made the goal one behaviour, buying from a second brand. The economics showed why a flat rate fails (₹1,500 crore, needing 15% more sales) and a new-brand bonus works (₹340 crore, 3.4%). Earn rates set by each brand’s margin, with a settlement pool, keep the brands in. The North Star counts cross-brand members, a holdout proves the bonus caused the extra buying, and the cost cap and consent checks are the guardrails.

FRAMEWORK · WORKING BACKWARDS: A PR-FAQ, THEN A PRE-MORTEM

1 · The situation. Everyone at Tata agrees “a group loyalty programme” is a good idea, and everyone pictures something different: a points bank, an app, a credit card, a VIP tier. Before building, the team needs one page that says what a customer gets, and a plain list of the ways it could fail.

2 · An everyday example. Planning a wedding. Before booking a hall, write the toast your friend will give on the day: “Everyone ate well, nobody waited, and the couple actually enjoyed it.” Then imagine it is the morning after and it went badly. Why? The caterer ran out, the band came late, the bride’s grandmother couldn’t climb the stairs. Now you know what to plan for.

	The wedding	The Tata programme
Press release	The toast, written first	A one-page announcement of the launch, for customers
Customer FAQ	“Is there parking? Veg food?”	“What is a coin worth? Where can I spend it?”
Internal FAQ	“Who pays the caterer? What if it rains?”	“Who funds a coin? How do old points move over?”
Pre-mortem	“It’s the morning after, and it went badly”	“It’s 2028, and the programme failed”

3 · The framework. Amazon plans new products by **working backwards** from the customer. The team writes a **PR-FAQ** before any building: a one-page press release announcing the finished product in plain words, then questions customers would ask, then harder internal questions about cost, risk and how it works. If the release isn’t exciting to a customer, the idea isn’t ready. A **pre-mortem**, from the psychologist Gary Klein, is the opposite check: the team imagines the launch has already failed and writes down why, then ranks the reasons (Figure 9.25). People who would never criticise a plan in a meeting will happily explain a failure that “already happened”.

THE PRESS RELEASE, ONE PARAGRAPH (WRITTEN FOR THIS EXERCISE; THE QUOTE IS INVENTED)

Tata Neu: every Tata brand now pays you back where you want it. Mumbai, June 2027. From today, every purchase at BigBasket, Croma, Westside, Tanishq, Titan, Taj, Air India and Starbucks earns NeuCoins, and every coin is worth ₹1 at all of them, at the counter or in the app, in any amount. Your first ₹10,000 at a Tata brand you haven't tried earns five times the usual coins. Your old points move over at their full rupee value, with a year to use them. "I earned enough on groceries this year to pay for my parents' airport lounge," says Meena Iyer, a teacher in Chennai. Join with your mobile number; you choose whether your brands can use your purchases to tailor offers.

Customer FAQ (sample): What is a coin worth? Where can I spend it? Do my old points expire? Can my family share one account? How do I stop brands sharing my data? · **Internal FAQ** (sample): Who funds a coin, and at what rate per brand? What does the programme cost a year, and what extra sales pay for it? How are old points valued? What stops fraud on recycled numbers? What does each brand's P&L look like after settlement?

Using it, step by step (on this case):

1. Write the press release for a customer, in one paragraph, with one quote from a real kind of person.
2. Write the five questions a customer would ask first, and answer them plainly.
3. Write the five internal questions a CFO or a brand head would ask; any you can't answer is the work plan.
4. Run the pre-mortem: "It's 2028 and it failed. Why?" Everyone writes reasons alone first, then share.
5. Score each reason for likelihood and damage, and fix the top three in the design before building.

COMMON SLIP · a press release full of features ("three tiers, dynamic offers, AI personalisation"). A customer reads for what she gets: "every coin is worth ₹1 everywhere".

Pre-mortem, "It's 2028 and the programme failed. Why?": causes ranked by likelihood × damage (illustrative scores)

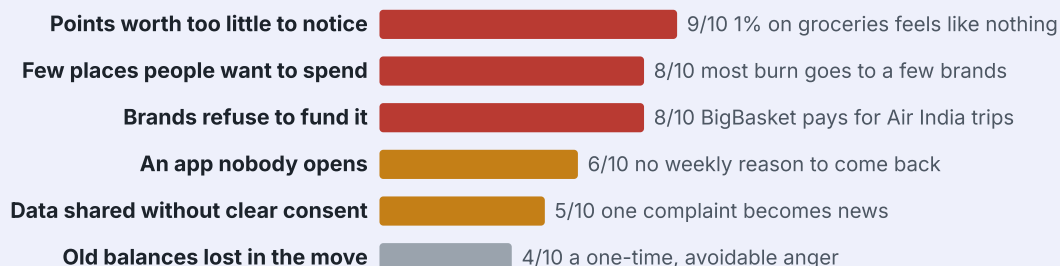


Figure 9.25 · A pre-mortem for the programme, ranked. The top three (coins worth too little, nowhere tempting to spend them, brands refusing to fund them) are all economics, which is why a good answer prices the coins before designing tiers. *Illustrative scores.*

4 · When it misleads. A PR-FAQ can become a writing contest: a beautiful page that nobody tests with a single customer. And a pre-mortem that lists twenty risks without ranking them leaves the team where it started, worried about everything. Neither replaces data: Tata Neu's own numbers on who earns and who spends are worth more than any imagined failure.

5 · In the interview. "Before designing the tiers, let me write the one-line promise a customer would read, that every coin is worth ₹1 at every Tata brand, and then imagine it's 2028 and it failed: the top reasons are all about what a coin is worth and who pays for it, so that's where I'll start."

CONSUMER HARDWARE, LOYALTY & PUBLIC SERVICES · GUESSTIMATES

Guesstimates: users, and orders per hour

Both guesstimates here count people and what they do, and both trip on a word in the question. "Users" can mean everyone who ever installed an app or the people who ordered this month, and the two differ by a factor of three or more. "Per hour" can mean an average hour or the busiest one, and for food delivery the dinner hour carries three times the average. The six cards still do the rest.

1

Pin down the unit

Monthly users of this app, or everyone who ever ordered? An average hour, or the peak?

2

Pick the approach

From people down (demand), or from riders and kitchens (supply)?

3

Draw the tree

People → who can order → who do → how often → this app's share.

4

Round numbers, one reason each

Why 35% order monthly? Why two orders each?

5

Check it another way

A company's published users or orders, or a similar country per head.

6

Give a range, and what moves it

Usually how often people order, and how sharp the peak is.

KEY IDEA · USERS ARE NOT ACTIVE USERS, AND AN HOUR IS NOT A DAY ÷ 24

An app with 5 crore downloads may have 2 crore people who ordered this year and 1 crore who ordered this month. Companies report the last number (monthly transacting users) because it is the one that pays. The same care applies to time. People eat at lunch and dinner, not at 4 am, so a day's orders divided by 24 describes an hour that never happens. Say which you are estimating: the average (for a yearly plan) or the peak (for how many riders and servers you need at 9 pm).

Try it first. Of ten adults you know in a big city, how many ordered food on an app in the last month? Would that be true in Lima or Bogotá? Build the number of users of one app in a country of 10 crore, for five minutes.

THE QUESTION

Estimate the number of users for a food delivery app in a South American country.

A Sure, I have a couple of clarifying questions before I start the estimation. What is the total population of the country, and should I assume this is a product in its early stage or already launched?

Q The country has a population of 100 million. You can assume the platform is already launched and is looking to scale. Go ahead with the assumptions you need.

A I'll assume the app has achieved some market penetration and is targeting urban consumers first, as they have better access to infrastructure, smartphones and restaurants. I'll also assume a significant share of users are urban working professionals or students, who drive demand for food delivery.

Q Seems logical. You can proceed with this assumption.

A Thank you. Before I begin with the calculation, I'd like to outline my segmentation framework:

1. **Urban vs rural:** given infrastructure and internet constraints in rural areas, adoption will concentrate in cities.
2. **Age:** those aged 18-40 are the most likely users, given lifestyle and disposable income.
3. **Income and digital access:** smartphone ownership, internet and digital payments as enablers.
4. **Food-delivery propensity:** the share of the digitally-enabled group likely to use delivery.

Are these factors exhaustive?

[The casebook prints this list, and the note on the 41+ group, as layout tables that split the sentences; both are set out here in reading order, in the casebook's words.]

Q That approach works. Assume the slicing percentages as you see fit.

A Great. I'll proceed with these assumptions:

1. **Urban population:** Assume 70% of 100M = 70 million
2. Age Distribution in urban areas: i. 0-17: 25% ii. 18-40: 50% iii. 41+: 25%

So, 18-40 urban population = 50% of 70M = 35 million

(I'm treating 18-40 as the core users; the 41+ group does order delivery too, at a lower rate, so this funnel is slightly conservative.)

3. Income and Digital Access (smartphone + internet + digital payments): Among 18-40 urban population: Assume 70% have sufficient income and access → $0.7 \times 35M = 24.5$ million

4. **Food Delivery Propensity:** Of the digitally active population, estimate 40% actively use food delivery platforms (based on competitive landscape and market maturity) → $0.4 \times 24.5M = 9.8$ million

Final Estimate: ~9.8 million active users

Q That figure seems reasonable. Can you explain your rationale for the 40% adoption assumption?

A Certainly. I based this on a few considerations:

1. **Market maturity:** Since the app is already launched and we're looking at a South American country with urban penetration, early adopters would already be on the platform.
2. **Competition:** Multiple food delivery platforms may exist, so active usage may be distributed across providers, reducing the app-specific share.
3. **User behavior:** While many users have access to the service, not all order frequently-some prefer cooking or dining out. I chose 40% as a mid-point assumption considering these factors.

Q Makes sense. We can close the discussion.

WHAT WORKED · WHAT WAS MISSING

- ✓ **Asked the population and the product's stage** before starting.
- ✓ **A clean funnel:** urban → age → access → propensity, each step multiplied.
- ✓ **Named the 41+ group it left out** and said the answer is on the low side for that reason.
- ✓ **Defended the 40%** with three reasons when asked.

- ✗ **"Users" was never defined.** Ever installed, ordered this year, or ordered this month? Each differs by a large factor.
- ✗ **The market and the app were mixed up.** 40% "use food-delivery platforms" counts all apps; the candidate then said competition lowers this app's share, but never applied a share.
- ✗ **70% of urban 18-40-year-olds with income and access** is generous for much of South America, where many pay in cash.
- ✗ **No check.** Brazil's iFood publishes orders and customers, a ready comparison per head.

Guesstimate kit check: 1 Unit ✗ **"users" undefined** · 2 Approach ✓ **demand side** · 3 Tree ✓ · 4 Numbers with reasons ✓ **for the 40%, when asked** · 5 Check ✗ · 6 Range ✗. Card 1 cost the most: without a unit, 9.8 million could be the whole market or one app, and nobody can tell whether it is right.

From people to this app's monthly users (illustrative)

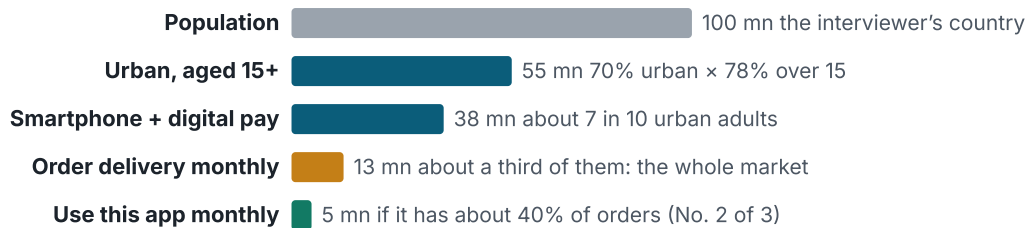


Figure 9.26 · From people to one app's monthly users. The orange bar is the whole market; the green bar is this app. The casebook's 9.8 million sits between them, because it counts the market but calls it the app. *Illustrative rates.*

STRONGER ANSWER



Figure 9.27 · The six cards, applied to app users. Card 1 (which users) and card 5 (Brazil per head) change the answer most. *Author's summary.*

- A 1 · UNIT**
I'll estimate monthly users of this app: people who ordered at least once in the last 30 days. That is what delivery companies report, and it is the number that brings revenue. I'll give the whole market's monthly orderers on the way.
Why this move: "users" is the vaguest word in product numbers. Picking the reported definition makes the answer checkable.
- A 2 · APPROACH**
Demand side, from people to orderers, then this app's share. I'll take all adults over 15 in cities rather than only 18–40, and apply one rate for how many order.
Why this move: a wider age band with one ordering rate is simpler than a narrow band plus a note that it leaves people out.
- A 3–4 · TREE AND NUMBERS**
10 crore people; about 70% live in cities, as in most of South America, and about 78% of them are over 15: 5.5 crore. About 70% have a smartphone and a way to pay online: 3.8 crore. About a third of those order food at least once a month: 1.3 crore, the whole market. If this app is second of three, with about 40% of orders: **about 50 lakh monthly users** (Figure 9.28).
Why this move: every step has a reason a listener can argue with, and the market and the app are separate numbers, so a wrong market share doesn't hide inside another rate.
- A 5 · CHECK**
Brazil's iFood handles about 12 crore orders a month for about 21 crore people: 0.57 orders a person a month. My market: 1.3 crore people × about 2 orders ÷ 10 crore = 0.26, about half of Brazil's. Brazil is the region's most developed delivery market, so half sounds right for a typical country.
Why this move: orders per person lets you compare countries of any size, and a published leader gives an upper bound.
- A 6 · RANGE**
3 to 8 million monthly users of this app, most likely about 5 million; the whole market 10 to 20 million. The biggest swing is how often people order: in a Brazil-like country the market would be nearer 20 million.
Why this move: a range with its swing factor tells the listener which assumption to test first.

HOW THE ANSWER HANGS TOGETHER

Defining users as monthly users of this app fixed what was being counted. The tree kept the market (1.3 crore) and the app (50 lakh) apart, so the share was a visible choice. Brazil's orders per person confirmed the market is a sensible size, and the range names ordering frequency as the number to check.

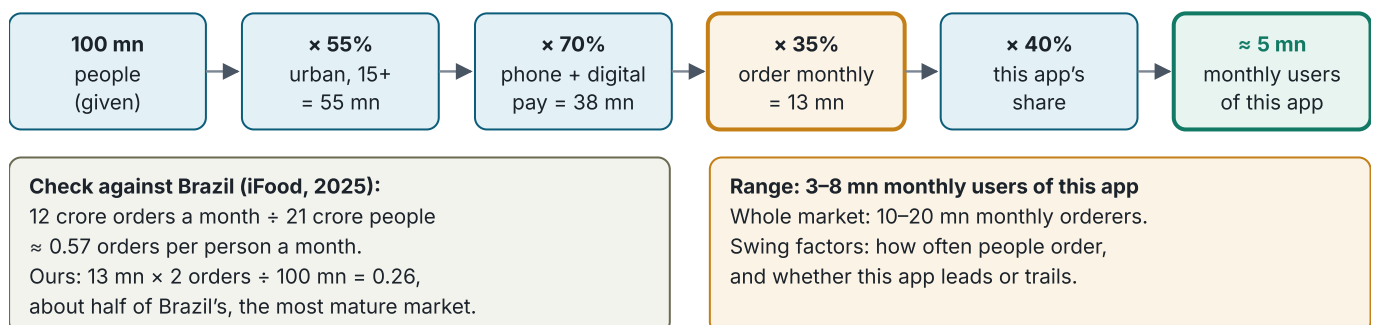


Figure 9.28 · Food-delivery users, as an equation tree. The orange box (how many order monthly) is the biggest swing factor; the check compares orders per person with Brazil. *iFood figures from 2025 press coverage; other rates illustrative.*

THE REUSABLE TRICK

Estimate the market first, then apply the company's share as a separate, named step. When a question says "users of an app", a number that skips the share is the market wearing the app's name.

GUESSTIMATE 9.2 · SWIGGY ORDERS PER HOUR IN INDIA

Guesstimate Easy IIM B Sigma PM Casebook 2026–27, p. 142

Try it first. When did you or your family last order food on an app, and at what time? How many times a month? Build Swiggy's orders in an hour, and say which hour, for five minutes.

THE QUESTION

Can you estimate the number of orders delivered by Swiggy per hour in India?

A Sure. I would like to clarify a few things before I begin. Should I consider both peak and off-peak hours in my estimation, or focus on an average over the entire day?"

Q You can decide.

A First, I'll estimate the number of food delivery app users in India, then determine the average number of orders per user per month, and finally apply Swiggy's market share. Does that sound good?

Q That is a good approach. Go ahead.

A I want to divide the population into 3 segments. Segment 1 having Tier-1 cities of population over 10 million, segment 2 having Tier-2 cities of population over 5 million and segment 3 consisting of Tier-3 cities and towns

Q Sure, please proceed.

A For the top 1-5 cities, I am assuming a penetration rate of 40% for people who order food online. Therefore, 10 million x 5 x 0.4 = 20 million users

For the top 6-10 cities, I am assuming a penetration rate of 30%. Therefore: 5 million x 5 x 0.3 = 7.5 million users

Adding another 5 million users from smaller cities, the total is: 20 + 7.5 + 5 = 32.5 million app users

To estimate the average number of orders per month, I want to divide the usage patterns into four categories: monthly, bi-monthly, weekly and alternate day. Is it okay?

Q Yes, that is a fair segmentation. You can proceed.

A Average users of food delivery apps:

Monthly : 40%

Bi-monthly: 30%

Weekly: 20%

Alternate day: 10%

Average number of orders per customer per month:

$$0.4 * 1 + 0.3 * 2 + 0.2 * 4 + 0.1 * 15 = 3.3$$

Total number of orders per month:

$$32.5 \text{ million} * 3.3 \sim 105 \text{ million orders}$$

Number of orders per day: 105 million / 30 ~ 3.5 million orders

Assuming Swiggy has a market share of 50%, number of Swiggy orders per day:

$$0.5 * 3.5 = 1.75 \text{ million orders per day}$$

Number of Swiggy orders per hour in India:

$$1.75 \text{ million} / 24 = 70,00 \text{ orders per hour (approx.)}$$

Q Seems practical. That would be it. Thank you.

WHAT WORKED · WHAT WAS MISSING

- ✓ **The right first question:** peak and off-peak, or an average?
- ✓ **A clear plan:** users → orders per user → Swiggy's share, agreed with the interviewer.
- ✓ **Usage in four frequency groups**, not one average.

- ✗ **Asked about the peak, then divided by 24** without saying so. The dinner hour carries about three times the average.
- ✗ **An arithmetic slip:** 1.75 million ÷ 24 is about 73,000, printed as "70,00".
- ✗ **Orders per user too high:** the alternate-day group alone adds 1.5 orders a month. Swiggy's own figures give about 2 orders per monthly user.
- ✗ **No check** against Swiggy's results, which publish food orders and monthly users every quarter.

Guesstimate kit check: 1 Unit ✗ **asked about the peak, then didn't choose** · 2 Approach ✓ **demand side** · 3 Tree ✓ · 4 Numbers with reasons ✗ **frequencies not explained** · 5 Check ✗ · 6 Range ✗. Card 1 cost the most: the one hour a delivery company plans for is 9 pm, and it is three times the answer given.

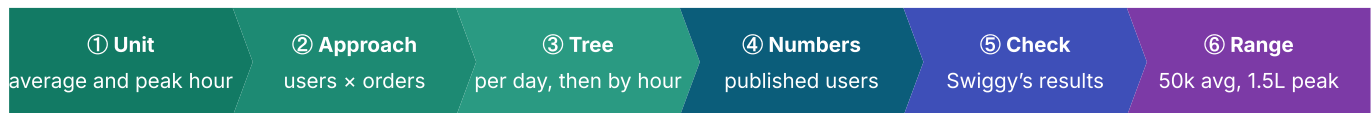


Figure 9.29 · The six cards, applied to orders per hour. Card 1 (average or peak) and card 5 (Swiggy's results) change the answer most. Author's summary.

A 1 · UNIT

Food orders delivered by Swiggy in India, not Instamart groceries. I'll give two numbers: the average hour, for a yearly plan, and the dinner peak, because that is the hour riders and servers must be planned for.

Why this move: answering both removes the guess about what the interviewer wanted, and the peak is the number an operations team actually uses.

A 2 · APPROACH

Demand side, but starting from Swiggy's own monthly users rather than India's population. Swiggy reports about 1.9 crore monthly food users (April–June 2026), so the market-share step goes away.

Why this move: a published user count replaces the two shakiest steps, city penetration and market share.

A 3–4 · TREE AND NUMBERS

A monthly user orders about twice a month: many order once, a few order every few days. $1.9 \text{ crore} \times 2 = 3.8 \text{ crore}$ orders a month $\div 30 =$ about **12.7 lakh a day**. $\div 24 =$ **53,000 in an average hour**. But orders follow meals (Figure 9.30): the 9–10 pm hour carries about 13% of the day, so the peak is $12.7 \text{ lakh} \times 13\% \approx$ **1.6 lakh orders**, three times the average.

Why this move: a day's total spread over an hourly shape gives both numbers from one tree, and makes the peak-to-average ratio visible.

Swiggy food orders in each hour of an average day, India (hour-of-day split illustrative; daily total from results)

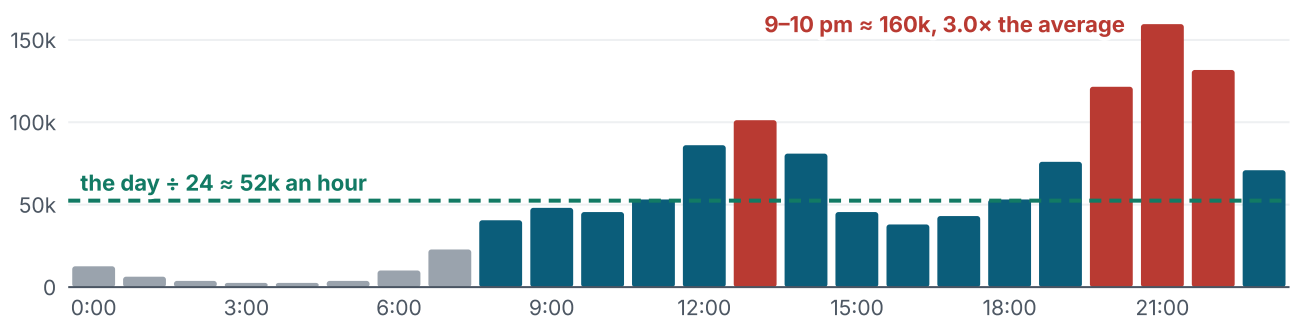


Figure 9.30 · Orders by hour of day. The dashed line is the day $\div 24$; real demand sits under it for 15 hours and far over it at lunch and dinner. Riders, kitchens and servers are sized for the red bars. Daily total from Swiggy's Apr–Jun 2026 results; the hourly shape is illustrative.

A 5 · CHECK

Swiggy delivered 11.45 crore food orders in April–June 2026: over 91 days that is 12.6 lakh a day, or 52,000 an hour on average. My tree gives 53,000. The casebook's 73,000 is about 40% too high, mostly because of the orders per user.

Why this move: a listed company's quarterly orders is the strongest check there is; it confirms the tree and shows where the casebook drifted.

A 6 · RANGE

45,000–60,000 orders in an average hour; 1.2–1.8 lakh in the dinner peak. The peak moves most with rain, cricket finals and festivals, when a single hour can go well past 2 lakh.

Why this move: naming what spikes the peak tells an operations team when to plan extra riders.

HOW THE ANSWER HANGS TOGETHER

Answering both the average and the peak settled the unit. Swiggy's published users removed the penetration and share guesses, two orders a month gave 12.7 lakh a day, and the hourly shape turned that into 53,000 average and 1.6 lakh at 9 pm. Swiggy's quarterly orders confirmed the day, and the range says what makes the peak jump.

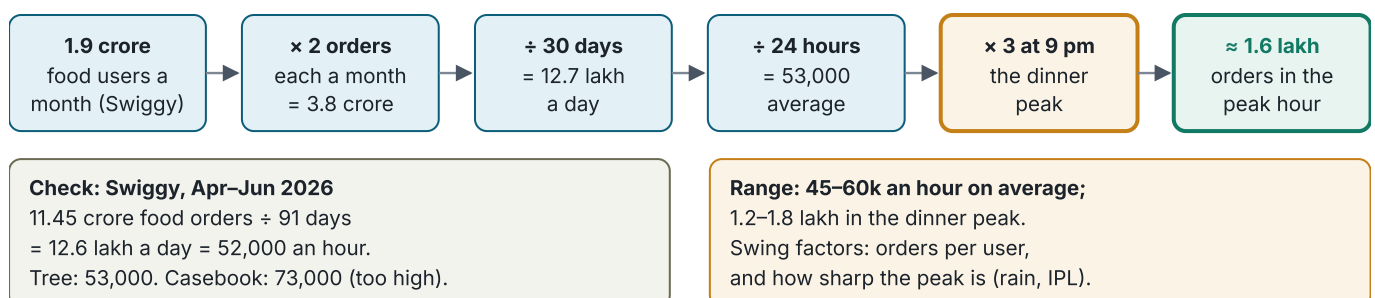


Figure 9.31 · Swiggy's orders per hour, as an equation tree. The orange box (how sharp the dinner peak is) is the biggest swing factor for planning; the check uses Swiggy's own quarterly orders. Swiggy Q1 FY27 results (Jul 2026); orders per user and peak share illustrative.

THE REUSABLE TRICK

For anything people do at set times (meals, commutes, cricket, payday), estimate the day, then ask what share falls in the busiest hour. Peak $\div$ average of 2.5–3 is typical for food delivery; the peak is what you build for.

Intuit: a deep dive into TurboTax, QuickBooks, Enterprise Suite, Credit Karma and Mailchimp

Intuit sells software for the money chores people and small firms dread: the yearly tax return, the books, invoices, payroll, a loan. It is a useful company for a PM to study because it has every model on one balance sheet: a once-a-year consumer product (TurboTax), a monthly subscription for small businesses (QuickBooks), a free app paid for by lenders (Credit Karma), a freemium marketing tool (Mailchimp), and now a sales-led product for mid-sized firms (Intuit Enterprise Suite). It also has a lesson for India: QuickBooks left the Indian market in 2023.

\$21.4 bn revenue in FY2026 (year to 31 Jul 2026), up 14%	42% operating margin before share-based pay and one-offs (\$8.9 bn); 27.5% as reported	3.9 crore US TurboTax returns in FY26, down 2%, while revenue rose 7%	+28% growth in mid-market customers (QuickBooks Advanced and IES) in FY26
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The company on one page

Scott Cook and Tom Proulx started Intuit in 1983 with Quicken, a program for balancing a household cheque book. Cook's idea was to watch people doing their finances at home and remove every step they found painful; Intuit still sends staff to sit with customers ("follow-me-home" visits). QuickBooks followed for small businesses, and in 1993 Intuit bought the maker of TurboTax. The big bets since then (Figure 9.32) have moved it from software people install to money it helps move: payments, payroll, loans, and lender fees.



Figure 9.32 · Intuit's big bets. Each purchase added a new way to earn from the same customers: lender fees (Credit Karma), marketing spend (Mailchimp), bigger contracts (Enterprise Suite). *Intuit filings and announcements; purchase prices as announced.*

Who it serves. Three groups, each with its own products (Figure 9.33). **Consumers** use TurboTax once a year and Credit Karma to watch their credit. **Small businesses**, from a freelance designer to a bakery with six staff, use QuickBooks and Mailchimp. **Mid-sized firms** with several companies or locations use Intuit Enterprise Suite. Around them sit **accountants**: they file taxes for clients with Intuit's professional software (ProTax), and they recommend QuickBooks to the small businesses they serve, which makes them Intuit's most important sales channel.

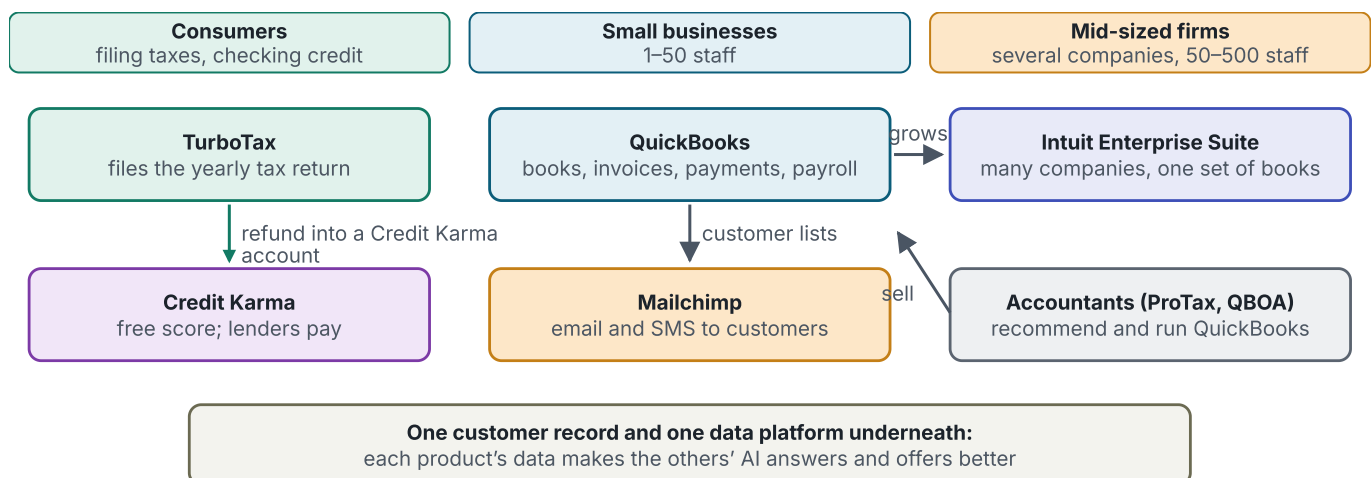


Figure 9.33 · How Intuit's products connect. Data and customers flow between products: a TurboTax refund can land in a Credit Karma account; QuickBooks customer lists feed Mailchimp; a growing QuickBooks firm moves up to Enterprise Suite. *Author's diagram from Intuit's product pages (2026).*

Where the money comes from. In FY2026 Intuit earned \$21.4 billion (Figure 9.34). Global Business Solutions (QuickBooks, Enterprise Suite, Mailchimp) brought \$12.9 billion, up 16%; within it, the online products grew 19%, or 23% without Mailchimp. The Consumer side (TurboTax, Credit Karma, ProTax) brought \$8.6 billion, up 11%. The parts Intuit calls its "big bets" (mid-market, expert help, money products) grew 34% and are now 30% of revenue. For FY2027 it expects 9-10% growth, and it now reports Mailchimp on its own, expecting it to stay flat.

Intuit's \$21.4 bn of revenue in FY2026 (year to 31 Jul 2026), by product

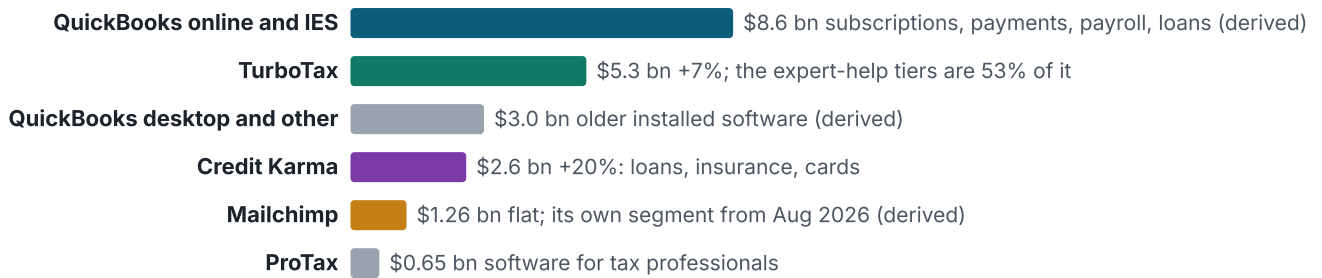


Figure 9.34 · Intuit's revenue by product, FY2026. QuickBooks and Enterprise Suite are the engine; TurboTax is large but slow; Credit Karma is the fastest-growing named business. *Intuit Q4 FY26 results (Aug 2026). QuickBooks online, desktop and Mailchimp split derived from reported segment totals and Mailchimp's FY27 guidance, so approximate.*

The data flywheel. Intuit's bet is that its data makes each product smarter (Figure 9.35). Every tax return teaches TurboTax which questions to ask; every categorised bank line teaches QuickBooks how to file the next one; every approved loan improves Credit Karma's approval odds. More data makes more of the work automatic, and automatic work lets Intuit sell "done for you" help (by an expert or an AI agent) at a higher price.

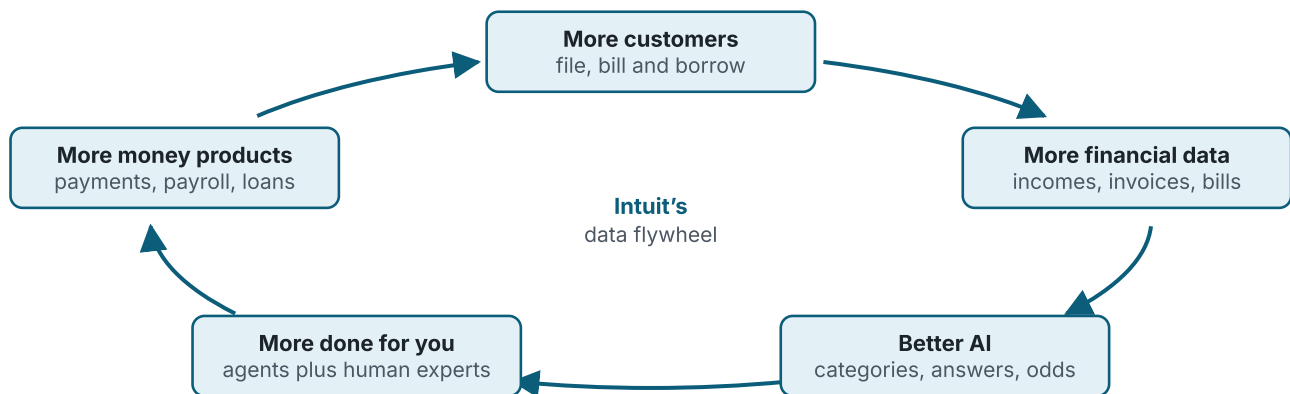
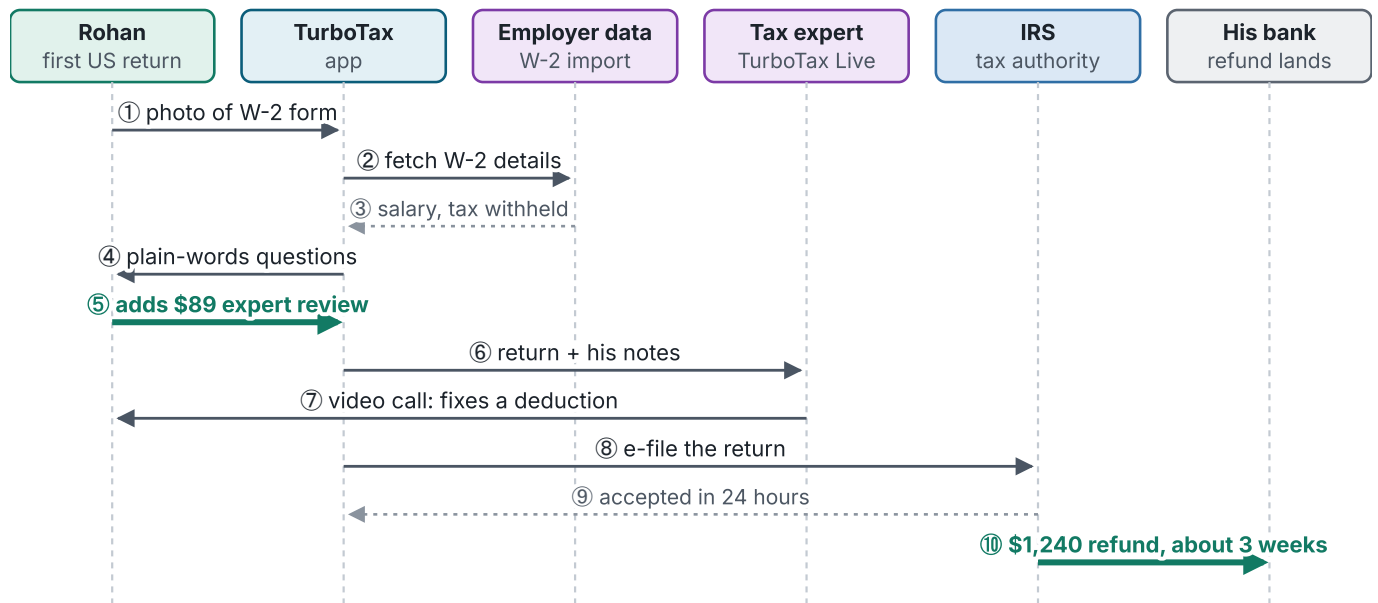


Figure 9.35 · The flywheel Intuit is betting on. It spins only while customers trust Intuit with their data, which is why a tax error or a misleading ad costs Intuit more than a normal software bug. *Author's diagram.*

TurboTax: the once-a-year product

Walkthrough (illustrative persona). Rohan, 29, moved from Pune to Austin for a software job and has to file his first US tax return. He opens TurboTax in February and takes a photo of his W-2 (the form from his employer showing salary and tax already taken out). TurboTax fills in the numbers and asks plain questions: did you move for work, do you have a bank account in India, did you sell any shares? He isn't sure how to report his Indian bank interest, so he pays \$89 to add a tax expert, who reviews his return on a video call and fixes a deduction. TurboTax sends the return to the IRS (the US tax department), which accepts it within a day, and his \$1,240 refund arrives about three weeks later (Figure 9.36).

Feature	User problem	Why it's designed this way	Metric it moves
Interview, not forms	"I don't understand the tax form"	Plain questions hide the form; the software places each answer	Returns completed; time to file
Data import	"Typing numbers is slow and error-prone"	W-2s, brokerage statements and last year's return pulled in	Errors; drop-off mid-return
Expert Assist	"What if I get it wrong?"	A tax expert on video checks and signs off, for a higher price	Revenue per return
Full Service	"Just do it for me"	Upload documents; an expert prepares and files the whole return	Revenue per return; share of "done for you"
Refund tracker	"When will my money come?"	Shows IRS status; offers to put the refund in Credit Karma Money	Credit Karma sign-ups



The expert step (5–7) is where TurboTax now earns most: 53% of its revenue in FY26.

Figure 9.36 · Behind a TurboTax return. Data is pulled in ((2–3)), the user answers questions instead of filling forms ((4)), and the paid expert step (5–7) checks the answer before filing. *Author's diagram; amounts and timings illustrative.*

Business model, with numbers. TurboTax earned \$5.3 billion in FY2026, up 7%, even though the number of US returns it filed fell 2% to 3.9 crore. The growth came from moving people up the price ladder (Figure 9.37): the expert-help tiers (TurboTax Live) grew 37% and are now 53% of TurboTax revenue. The money arrives in a rush: most returns are filed between February and April, so most of the year's TurboTax revenue lands in one quarter (Figure 9.38). That shapes everything, from hiring thousands of seasonal experts to freezing risky changes in January.

TurboTax's price ladder, federal return (illustrative US prices; state returns extra)

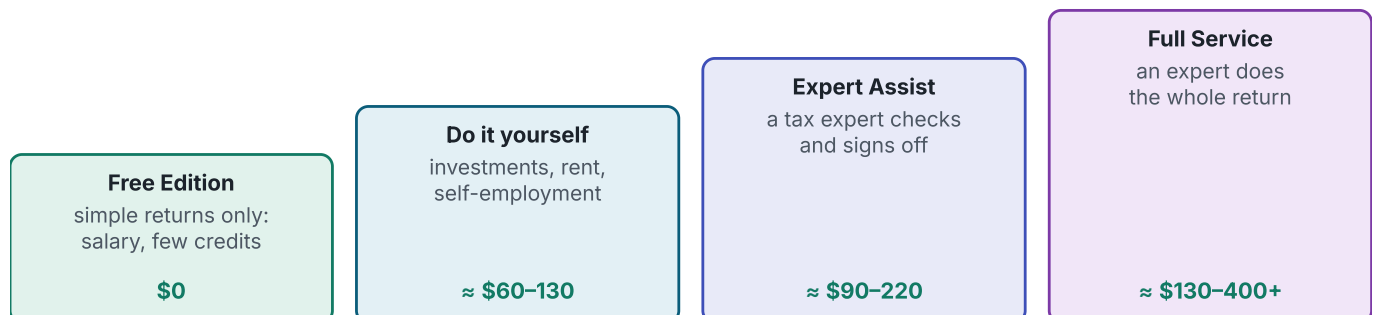


Figure 9.37 · The TurboTax price ladder. The free edition covers only simple returns; anyone with investments, rent or freelance income moves up, and each step up adds more human help. *Tiers from TurboTax's US site (2026); prices vary by year, return and offer, so shown as rough illustrative ranges.*

TurboTax revenue by quarter
share of the year (illustrative)

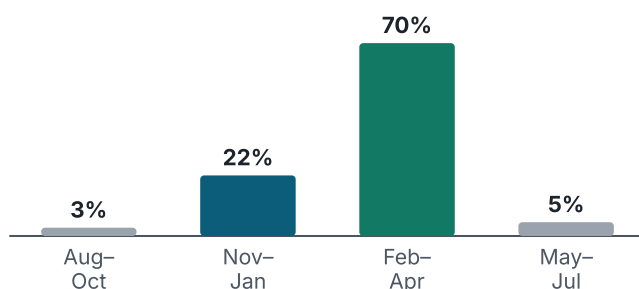


Figure 9.38 · One season pays for the year. Most returns are filed by the April deadline. *Illustrative split, consistent with Intuit's seasonal Consumer revenue.*

WHEN IT BREAKS · "FREE" THAT WASN'T

TurboTax advertised "free" filing for years while most filers didn't qualify for the free edition. In 2022 Intuit agreed to pay \$141 million to settle claims by all 50 US states, and in 2024 the US Federal Trade Commission ruled the ads deceptive. The IRS also tested its own free filing tool, Direct File, in 2024 and 2025. The lesson for a PM: a free tier that exists mainly to advertise erodes trust, and in tax, trust is the product.

NORTH STAR

Returns filed correctly and on time, weighted by the help level chosen

INPUTS

Share of data imported, not typed; completion rate; share moving to expert help; returning filers

GUARDRAILS

Amended returns and IRS notices; refunds for "free" disputes; expert wait times in April

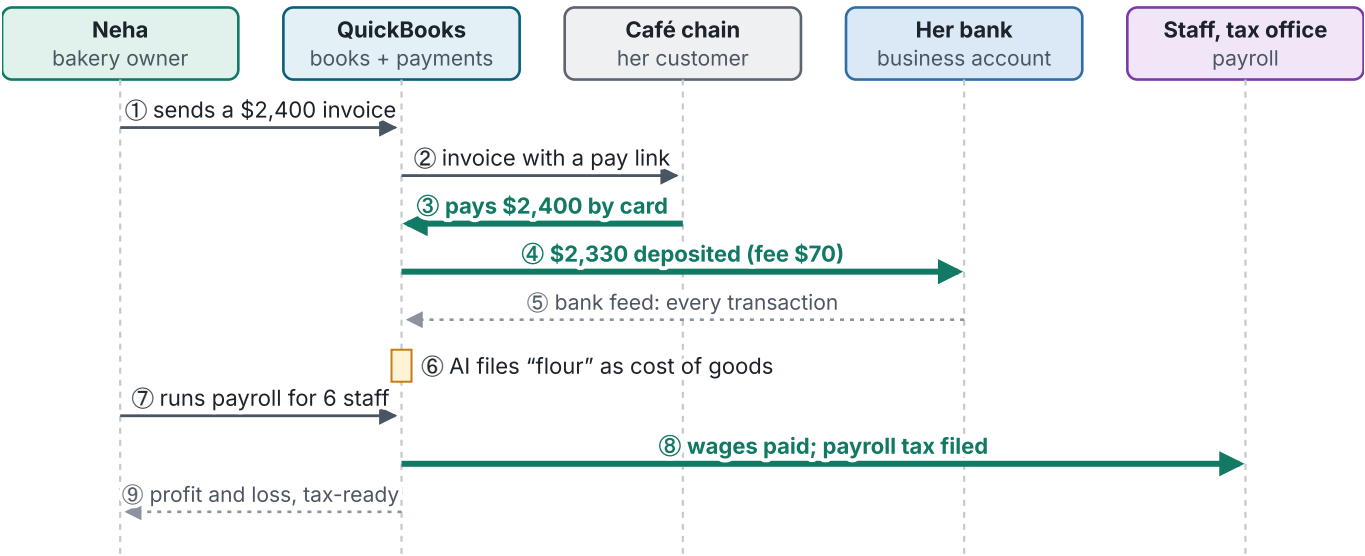
Moat, in plain words. Last year’s return is already inside TurboTax, so filing again there is easier than anywhere else; people are afraid of tax mistakes and stay with the brand they trust; and thousands of experts are hired and trained for one season, which a newcomer can’t match quickly. **Weakness:** the number of people filing with it is flat, so growth depends on selling more help to the same people.

Build next – Problem: people like Rohan, with income in two countries, are anxious and expensive to serve. **Bet:** an “income from abroad” path that imports foreign bank statements and explains tax treaties in plain words, with an expert check included. **Metric:** completion rate and amended returns for filers with foreign income.

Build next – Problem: the product is opened once a year, so nothing is learnt in between. **Bet:** a year-round “tax health” check in Credit Karma (withholding too high or too low, with a fix suggested in July). **Metric:** share of filers who adjust withholding; next year’s refund surprise.

QuickBooks: the books, and the money that moves through them

Walkthrough (illustrative persona). Neha runs a bakery in New Jersey with six staff and sells bread to a café chain. On Monday she sends the chain a \$2,400 invoice from QuickBooks with a “pay now” link. The café pays by card; QuickBooks Payments deposits \$2,330 in her bank the next day, keeping a fee. Her bank sends every transaction to QuickBooks automatically (the **bank feed**), and QuickBooks files the \$600 flour bill under “cost of goods” on its own, because it has seen thousands of bakeries’ flour bills. On Friday she runs payroll: wages go to the staff and payroll taxes are filed. At year end her accountant opens the same books and prepares her tax return (Figure 9.39).



The subscription is the door; payments (④) and payroll (⑧) earn as much again.

Figure 9.39 · Behind a week of a small business on QuickBooks. Money passes through Intuit twice (③–④ and ⑧), and each time Intuit earns. The categorisation at ⑥ is the AI that saves the most time. Author’s diagram; amounts and fee illustrative.

Feature	User problem	Why it’s designed this way	Metric it moves
Bank feeds	“I don’t have time to type every bill”	Transactions flow in daily; the owner only checks them	Weekly active businesses
Auto-categorisation	“I don’t know which account this goes in”	Learned from millions of businesses’ books; accepted with one tap	Share of lines filed without edits
Invoices with a pay link	“Customers pay late”	A card or bank payment in one tap; reminders sent automatically	Days to get paid; payments volume
Payroll	“Payroll tax rules scare me”	Wages, tax filing and year-end forms in the same place as the books	Payroll attach rate
Accountant access	“My accountant needs my books”	Free QuickBooks for accountants (QBOA); they manage many clients	New customers via accountants

Business model, with numbers. A subscription opens the door: plans run from about \$38 to \$275 a month in the US. The bigger money is in what moves through the books. A business on the Plus plan pays about \$1,380 a year; add card payments, payroll and a small loan, and the same business may pay Intuit nearly three times that (Figure 9.40). This is why Intuit’s online business revenue grew 19% in FY2026 while customer numbers grew more slowly: more money moves through each customer, and more customers buy payroll and payments. Mid-sized customers on the Advanced plan use payroll and payments more than smaller ones (13 and 9 points more, Intuit says).

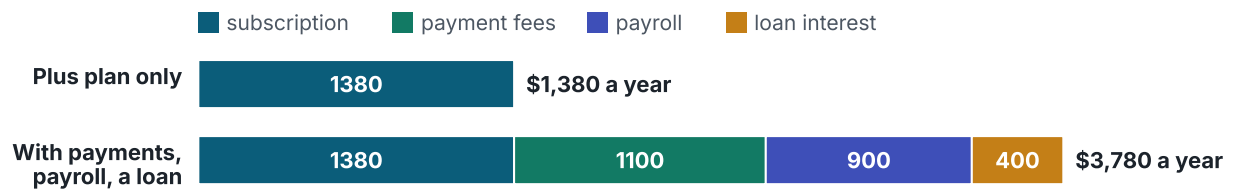


Figure 9.40 · What one small business pays Intuit in a year: the subscription is the smaller part. Payment fees, payroll and loan interest together can earn more than the software. Every extra service also makes leaving harder. *Plan price from QuickBooks' US site (2025–26, approximate); other amounts illustrative.*

Moat, in plain words. Years of books are hard to move: every invoice, bill, payroll run and tax filing lives inside. About the accountants: once an accountant knows QuickBooks and manages 60 clients in it, she recommends it to the next 60. And the more money a business moves through Intuit, the more reasons it has to stay. **Weaknesses:** the price rose steadily through 2023–26, and many small firms complain about it; and the software is built around US tax and payroll rules, which makes it hard to take to other countries.

REAL-LIFE EXAMPLE · WHY QUICKBOOKS LEFT INDIA

QuickBooks sold in India for over a decade, then stopped: new sales ended in 2022 and the service shut down for Indian customers in 2023, with about 30,000 users asked to move. Intuit gave no detailed reason, but the pattern is clear. Indian small businesses need GST invoices and returns, e-way bills and TDS, which change often; they pay little for software; and local products were built for exactly these rules. Tally has been the default for Indian accountants for decades, Zoho Books sells cheaply to firms already on Zoho, and Vyapar and Khatabook serve the smallest shops on a phone. Intuit kept its large development centre in Bengaluru, building for other markets. **The lesson:** in accounting, the local tax rules are the product, and the accountant who recommends the software is the channel. A global product without both loses to local ones.

NORTH STAR

Businesses that run their money through QuickBooks each month (books plus at least one money service)

INPUTS

Bank feeds connected; lines filed without edits; payments and payroll attach; accountant referrals

GUARDRAILS

Wrong categories that reach a tax return; churn after a price rise; payment disputes

Intuit Enterprise Suite: moving up to mid-sized firms

Walkthrough (illustrative persona). Neha's bakery grows. Five years on it is a group: the bakery company, a café company with four outlets, and a central kitchen company that makes dough for both. Each is a separate company for tax. On QuickBooks she runs three sets of books, and every month her finance manager spends four days copying numbers into a spreadsheet to see how the group is doing. Worse, the kitchen's sales of dough to the bakery and café count as revenue twice if the three sets of books are simply added up (Figure 9.41). This is the moment small-business software is outgrown.

A group of three companies: revenue for the year (\$ million, illustrative)

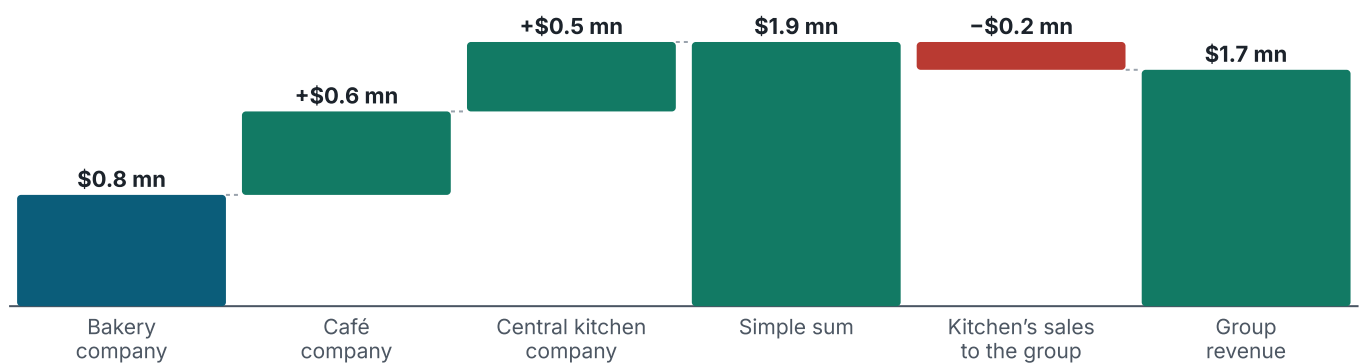


Figure 9.41 · Why a group needs consolidation. Adding the three companies gives \$1.9 million, but \$0.2 million is the kitchen selling to its own sister companies. Group revenue is \$1.7 million. Enterprise Suite removes such sales automatically. *Illustrative amounts.*

What it does. Intuit Enterprise Suite (IES), launched in September 2024, keeps the familiar QuickBooks screens but adds what a group needs. **Multi-entity accounting:** several companies in one login, with shared customers and suppliers. **Consolidation:** one group profit and loss, with sales between sister companies removed. **Dimensions:** every line can be tagged by outlet, project or product line, so the group sees which café makes money. On top come project costing, payroll and HR, Mailchimp marketing, and AI agents that do parts of the month-end work. The alternative for these firms used to be an **ERP** (enterprise resource planning software: one system for accounts, purchasing, stock and people) such as Oracle NetSuite, Sage Intacct or Microsoft Dynamics, which takes months to set up and costs far more.

QuickBooks plans, price a month (US list prices, approximate, 2025–26)

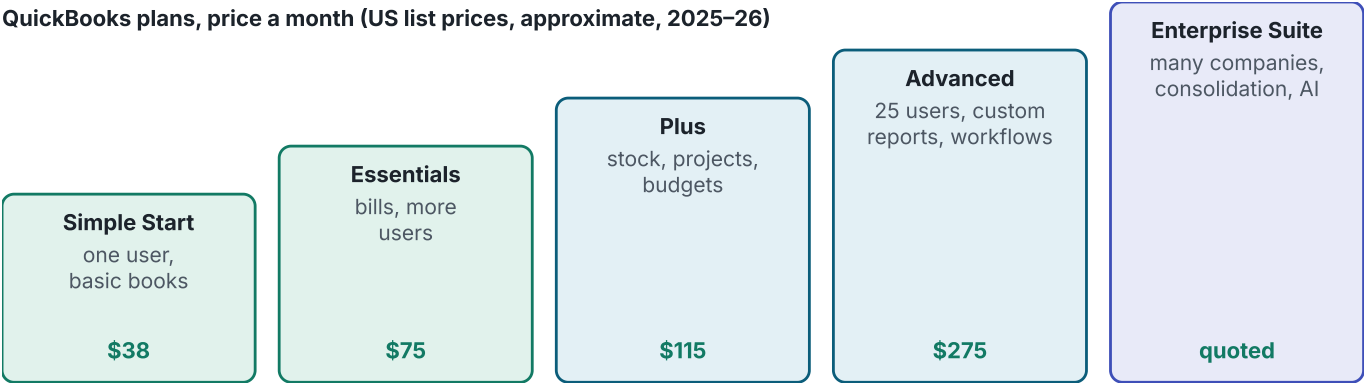


Figure 9.42 · The ladder a growing business climbs. Before IES, the step after Advanced was to leave for an ERP. IES keeps the customer, and its data, in-side Intuit. QuickBooks US list prices (approximate, 2025–26); IES is priced by quote.

Business model. IES is sold, not signed up for. Account managers and partner accountants run demos, move the data from QuickBooks or an old system, and sign yearly contracts worth many times a small-business plan. Intuit reported its mid-market customers (Advanced and IES) grew 28% in FY2026, and revenue from them 38% in the last quarter, the fastest-growing part of the company. The PM risks are those of any move up-market: bigger customers expect features, support and uptime a small-business product never needed; a failed migration can lose a customer for good; and a sales team is expensive, so contracts must be large enough to pay for it.

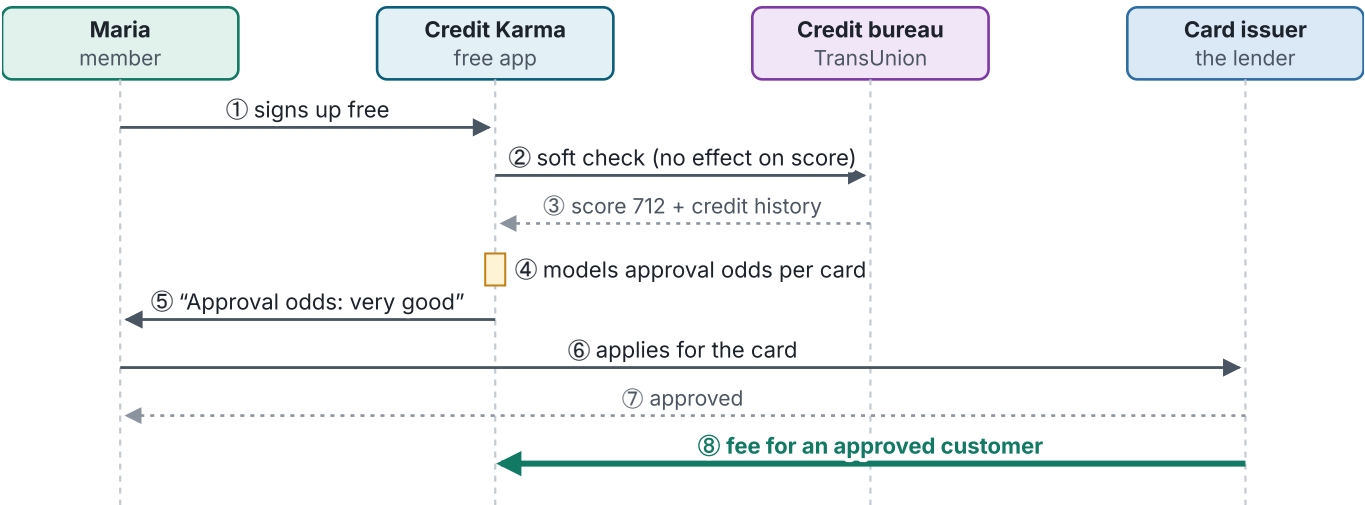
NORTH STAR Groups closing their month in IES (consolidated books signed off)	INPUTS Days to close the month; QuickBooks customers upgrading; migrations finished on time; modules per group	GUARDRAILS Migration failures; groups leaving for an ERP within two years; support tickets per customer
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Build next – Problem: a group’s finance manager spends days each month matching sales between sister companies. **Bet:** an agent that finds and matches those sales across companies and flags the ones that don’t agree, for a person to approve. **Metric:** days to close the month; share of matches approved without edits.

Build next – Problem: QuickBooks customers leave for an ERP before Intuit notices they have outgrown it. **Bet:** a signal (third company added, spreadsheets exported monthly) that triggers an IES offer through the customer’s accountant. **Metric:** upgrades vs losses to an ERP among customers with three or more companies.

Credit Karma: free to the user, paid for by lenders

Walkthrough (illustrative persona). Maria, 26, in Texas, wants her first rewards credit card but fears being turned down. She signs up to Credit Karma free. It checks her credit file with a bureau using a **soft check** (a look that doesn’t lower her score), shows her score of 712, and lists cards with her **approval odds**: “very good” for two, “fair” for a third. She applies for one marked “very good” and is approved. The card issuer pays Credit Karma a fee for an approved customer (Figure 9.43).



The member pays nothing; the lender pays when she is approved. Revenue swings with lenders’ appetite.

Figure 9.43 · Behind a Credit Karma card offer. The value to the member is fewer rejections (⑤); the value to the lender is a customer likely to be approved; the fee (⑧) is paid only on approval. Author’s diagram; score illustrative.

Business model. Credit Karma is a **lead-generation** business: lenders and insurers pay for customers who are approved or who buy. It earned \$2.6 billion in FY2026, up 20%, from personal loans, car insurance and credit cards. Its revenue swings with lenders’ appetite: when interest rates rose in 2022–23, lenders cut back and Credit Karma’s revenue fell, though nothing in the app had changed. Intuit ties it to TurboTax: filers can have their refund paid into a Credit Karma

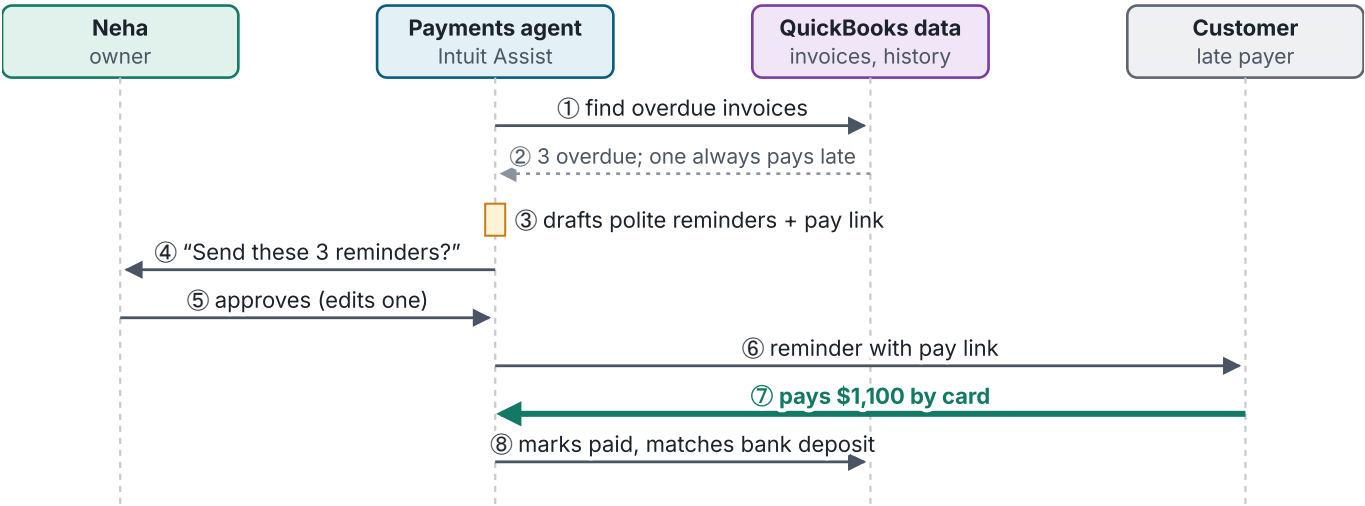
Money account, and TurboTax data can pre-fill Credit Karma’s view of a member’s income. **The PM tension:** the member trusts Credit Karma to show the best card for her; the money comes from lenders. If the ranking starts following fees instead of approval odds, members notice their rejections, and the trust that made the model work goes. The guardrail is the approval rate of offers marked “very good”.

Mailchimp: the purchase that stalled

Mailchimp lets a small business send emails and text messages to its customers: a bakery’s weekly specials, a shop’s sale. It is **freemium**: free for a small list, then priced by the number of contacts. Intuit paid about \$12 billion for it in 2021, when it earned about \$800 million a year and grew about 20%. The plan was to link it with QuickBooks: a business’s customer list and sales history would let Mailchimp send the right offer to the right customer, and show the sales it brought in. It hasn’t worked as planned. By FY2026 Mailchimp’s revenue was shrinking; Intuit now reports it separately from 1 August 2026 and expects it to stay flat at about \$1.26 billion. Small businesses moved to cheaper tools, to WhatsApp and Instagram for reaching customers, and to AI tools that write campaigns. **The lesson for a PM:** “their data plus our data” is a reason to buy a company only if customers actually use the two products together; count how many do before counting the synergy.

Intuit Assist and AI agents: from software to “done for you”

Intuit launched **Intuit Assist** in 2023, an AI assistant across its products, and in July 2025 a set of **AI agents** for QuickBooks: a payments agent that chases invoices, an accounting agent that categorises and reconciles transactions, a finance agent that prepares reports, and a customer agent that chases new leads. Here is the payments agent at work for Neha (Figure 9.44). It finds three overdue invoices, notices one customer always pays late, drafts polite reminders with a pay link, and asks Neha to approve. She edits one and sends. The customer pays by card; the agent marks the invoice paid and matches the bank deposit. Intuit reports that invoices chased this way are paid about five days sooner on average, and overdue ones are 10% more likely to be paid in full.



Intuit reports invoices chased this way are paid about 5 days sooner on average (2025).

Figure 9.44 · A payments agent chasing late invoices. The agent reads and drafts on its own (①–③) but asks before contacting customers (④–⑤), because a rude or wrong reminder costs the owner a customer. Author’s diagram; Intuit’s figure of about 5 days sooner (2025).

Experts plus AI. Intuit’s bigger idea is to combine agents with people: AI does the routine work, and a human book-keeper or tax expert reviews and signs off, sold as a higher tier. In tax and accounts a wrong answer has a legal cost, so the human check is part of what customers pay for. **Where it breaks:** an agent that files a transaction to the wrong account can put an error in a tax return months later, long after anyone would notice; an agent sending reminders on its own can annoy a customer the owner has known for years. **What it means for pricing:** if an agent does the bookkeeping, a price per user makes less sense than a price for work done, which is why Intuit bundles agents into higher plans and sells expert help per task.

Intuit in numbers: moat, metrics and moves

Moat, in plain words. Intuit holds the financial records of crores of households and small firms, and moving them is painful. Accountants learnt its software and recommend it. Its brand is trusted in the one area where people fear mistakes most. And each money product (payments, payroll, loans) adds another reason to stay. **2025–26 moves:** AI agents across QuickBooks (July 2025); Enterprise Suite pushed up-market, with mid-market customers up 28%; expert help grown to over half of TurboTax revenue; Mailchimp split out as its own segment (August 2026).

NORTH STAR Customers for whom Intuit does the money work (books, taxes or credit) each month	INPUTS Money products per customer; share of work done by agents; customers served by experts; upgrades to IES	GUARDRAILS Errors reaching a tax return; complaints about price rises or ads; approval rate of “very good” offers
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STRENGTHS

- ✓ The records of years of books and returns, which are hard to move
- ✓ Accountants as a channel and as a check on quality
- ✓ Money products that grow faster than subscriptions
- ✓ A path up-market (IES) that keeps growing customers

WEAKNESSES

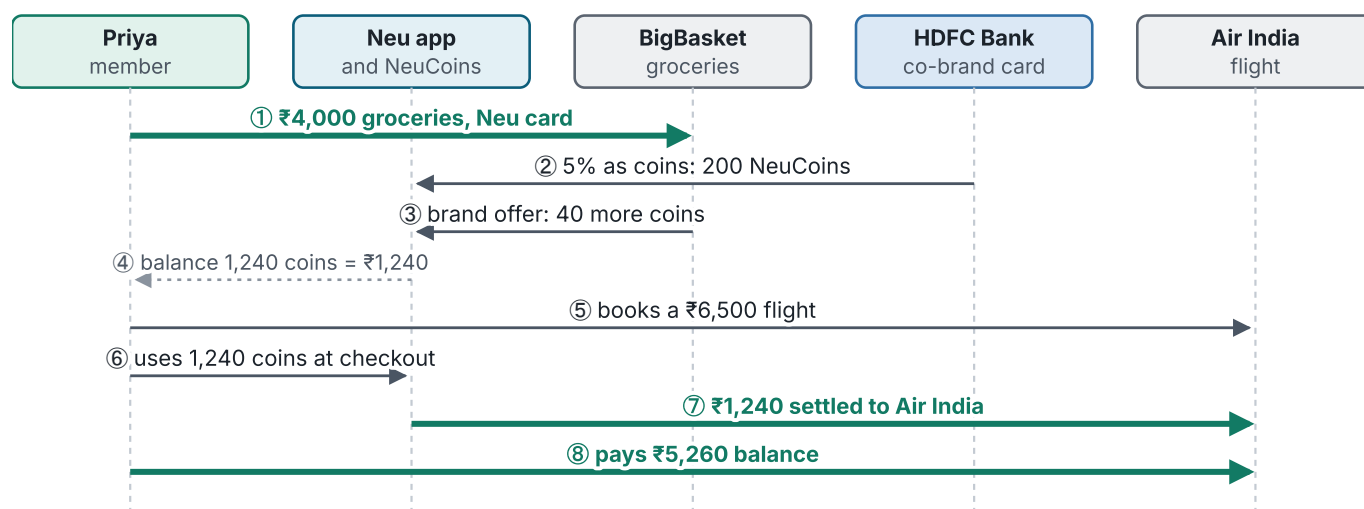
- TurboTax filers are flat; growth relies on selling more help
- Trust damaged by “free” advertising; price-rise complaints
- Credit Karma’s revenue depends on lenders’ mood
- Mailchimp shrinking after a \$12 bn purchase; exit from India

Build next → Problem: small firms learn they are short of cash too late. **Bet:** a 90-day cash forecast from QuickBooks data (invoices due, bills, payroll) that offers a small loan before the gap, priced from the firm’s own records. **Metric:** share of cash shortfalls flagged 30 days early; loan losses.

Build next → Problem: agents make mistakes that surface months later. **Bet:** a monthly “agent review” where the owner’s accountant sees every change an agent made, sorted by risk, and signs them off. **Metric:** share of agent changes reviewed; errors found at year end.

Tata Neu

Walkthrough (illustrative persona). Priya, 34, in Pune, pays for ₹4,000 of groceries on BigBasket with her Tata Neu HDFC Bank credit card. The card gives her 5% back as NeuCoins, 200 coins, and BigBasket adds a 40-coin offer. Over a few months her balance reaches 1,240 coins; each coin is worth ₹1. In October she books a ₹6,500 Air India flight inside the Neu app, spends all 1,240 coins at checkout and pays the rest by card. Air India is paid for the coins she spent from the programme’s pool (Figure 9.45).



Most coins come from the card (②), which HDFC pays for; brands add offers (③).

Figure 9.45 · Behind earning and spending NeuCoins. Most coins come from the co-branded card (②), paid for by the bank from its card fees; the brand adds offers (③); the brand where coins are spent is paid from the pool (⑦). *Author’s diagram; earn rates from HDFC Bank’s card page (up to 5% on Tata brands, 2025–26); amounts illustrative.*

What it is. Tata Neu launched in April 2022 as a “super-app”: one login for BigBasket, Croma, Tata CLiQ, Tata 1mg, Westside, Air India, Taj hotels, Titan and others, with NeuCoins as a shared currency and NeuPass as the loyalty membership. Tata Digital, which runs it, invested heavily and still loses money: ₹4,974 crore in FY26 on ₹35,990 crore of revenue, with BigBasket making up 64% of the loss.

What went wrong. A super-app needs a reason to open it every day, as WeChat has messaging and PhonePe has payments. Neu had none: people opened BigBasket for groceries and Air India for flights directly, and saw little reason to go through Neu. The brands underneath ran their own apps, prices and offers, so the experience felt stitched together. Coins were earned but often not spent, which made them feel worth less than ₹1.

What worked, and the change. The co-branded card did. It had passed 20 lakh cards by March 2025, took over 13% of new credit cards issued in one quarter, and carried more than ₹800 crore a month of UPI spends. Card users earn coins everywhere, so the coins build up fast enough to matter. By 2025–26 Tata Digital had refocused Neu on financial services (the card, payments, loans and insurance) and loyalty, rather than on being a shopping mall.

NORTH STAR

Members who earn and spend coins across two or more brands in 90 days

INPUTS

Active cards; coins earned per member; coins spent ÷ coins earned; first purchase at a new brand

GUARDRAILS

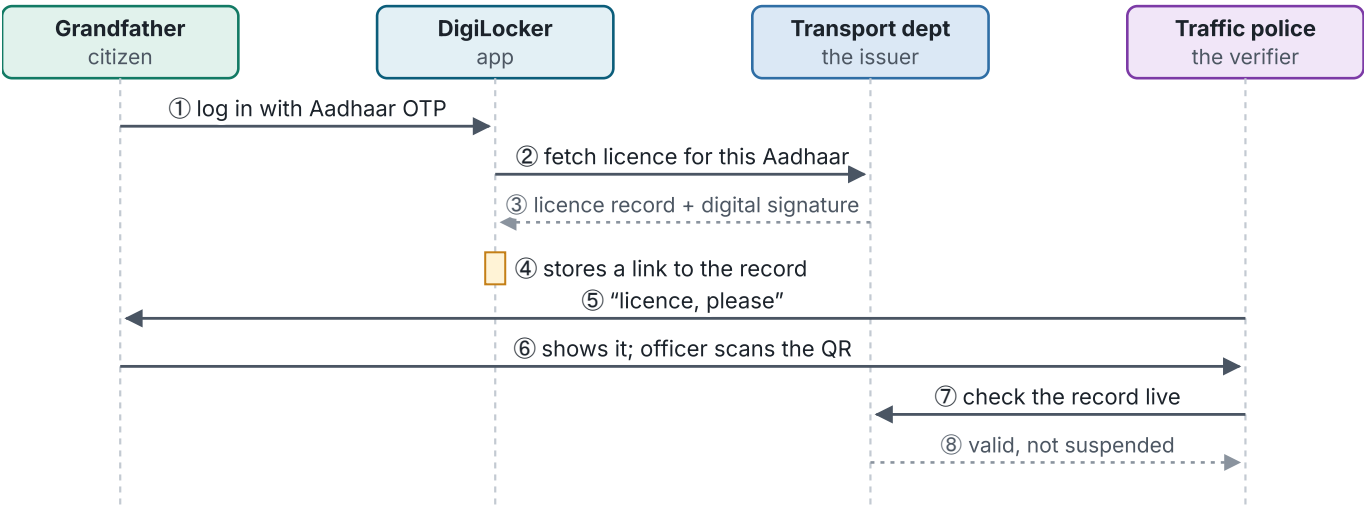
Cost of coins as a share of sales; card defaults; complaints about missing coins

Moat, in plain words. Trusted brands that people already buy from, and a card that turns everyday spending into coins for them. **Weakness:** no daily habit of its own, and brands that answer to their own profit targets.

Build next → Problem: coins sit unspent and feel worthless. **Bet:** “pay with coins” at the counter of every Tata store and café, suggested automatically when the balance covers the bill. **Metric:** coins spent ÷ coins earned; members spending at a second brand.

DigiLocker (with ONDC for comparison)

Walkthrough (illustrative persona). Ananya’s grandfather, 71, is stopped at a traffic check in Lucknow without his licence. He opens DigiLocker, which he set up last year with an Aadhaar OTP; the app had then fetched his driving licence straight from the transport department’s records. He shows it; the constable scans its QR code with the police app, which checks the record live: valid, not suspended. He is on his way in two minutes (Figure 9.46).



The issuer, not the citizen, vouches for the document; nothing is scanned or photocopied.

Figure 9.46 · Behind showing a licence from DigiLocker. The document comes from the issuer ((2)–(3)), and the verifier checks with the issuer too ((7)–(8)). The phone is only the window. *Author’s diagram based on DigiLocker’s documented flow (2026).*

How it works, and why it is trusted. DigiLocker, launched in 2015 by the government’s digital-services arm, holds two kinds of document (Figure 9.47). **Issued documents** are fetched from the body that issued them (CBSE mark sheets, driving licences, vehicle registrations, PAN, insurance policies), carry its digital signature, and under the IT rules count the same as the original. **Uploaded documents** are scans a citizen adds for their own use, with no special standing. By March 2026 DigiLocker had 67.6 crore users, and over 950 crore documents had been issued through it. Banks use it for KYC, universities for admissions, and employers to check degrees, each with the citizen’s consent.

Uploaded copy	Issued document
A photo of a paper licence anyone can edit a photo may be out of date or suspended officer must trust the citizen like a photocopy: needs attestation	A record fetched from the issuer carries the issuer’s digital signature checked live: current status officer trusts the transport dept legally equal to the original

Figure 9.47 · Uploaded copy vs issued document. The value is in the right-hand box: the issuer vouches for the document, so the person checking it doesn’t have to trust the person showing it. *Author’s comparison; DigiLocker (2026).*

Business model. None in the usual sense: the state pays, and the return is fewer office visits, fewer forged certificates and faster checks. The costs a PM watches are uptime at peak times (results day for board exams), how many issuers are connected, and help for people who get locked out when their Aadhaar-linked number changes. **ONDC, for comparison,** is the same idea applied to shopping: an open network (launched 2022) where any buyer app can show any seller’s goods, so a kirana shop on one app can sell to a customer on another. Where DigiLocker succeeded because a government department is the only issuer of a licence, ONDC must persuade private buyers and sellers to join, and orders have been harder to sustain once early discounts ended.

NORTH STAR	INPUTS	GUARDRAILS
Checks done with an issued document instead of a paper copy, each month	Issuers connected; documents issued; verifiers (banks, police, colleges) accepting it; sign-ups that fetch a document	Uptime on results day; wrong or stale records; access requests without consent

Build next – Problem: many of the elderly can’t manage the app when a document is needed. **Bet:** a “family helper” mode where a child can show a parent’s document, with the parent’s one-time consent and a log the parent can see. **Metric:** checks completed for users over 60; consent revocations.

Side by side

TurboTax	QuickBooks	IES	Credit Karma	Tata Neu	DigiLocker
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Job	File the yearly tax return	Books, invoices, payroll	One set of books for a group	Know your credit; get approved	One currency across Tata brands	Show a document the issuer vouches for
Who pays	The filer	The business, plus fees on money moved	The group, by contract	Lenders and insurers	The card bank and brands	The state
How often used	Once a year	Daily or weekly	Daily	Monthly	Weekly, if it works	A few times a year
Latest number	\$5.3 bn, +7% (FY26)	Online +19% (FY26)	Mid-market customers +28%	\$2.6 bn, +20%	20 lakh+ cards; ₹4,974 Cr loss (FY26)	67.6 crore users
Moat	Last year's return inside	Years of books; accountants	Grows from QuickBooks	Approval-odds data	Brands and the card	The issuers' signatures
Biggest threat	Free government filing; trust	Price rises; local rivals	ERPs; hard migrations	Lenders pulling back	No daily habit	Outages; exclusion

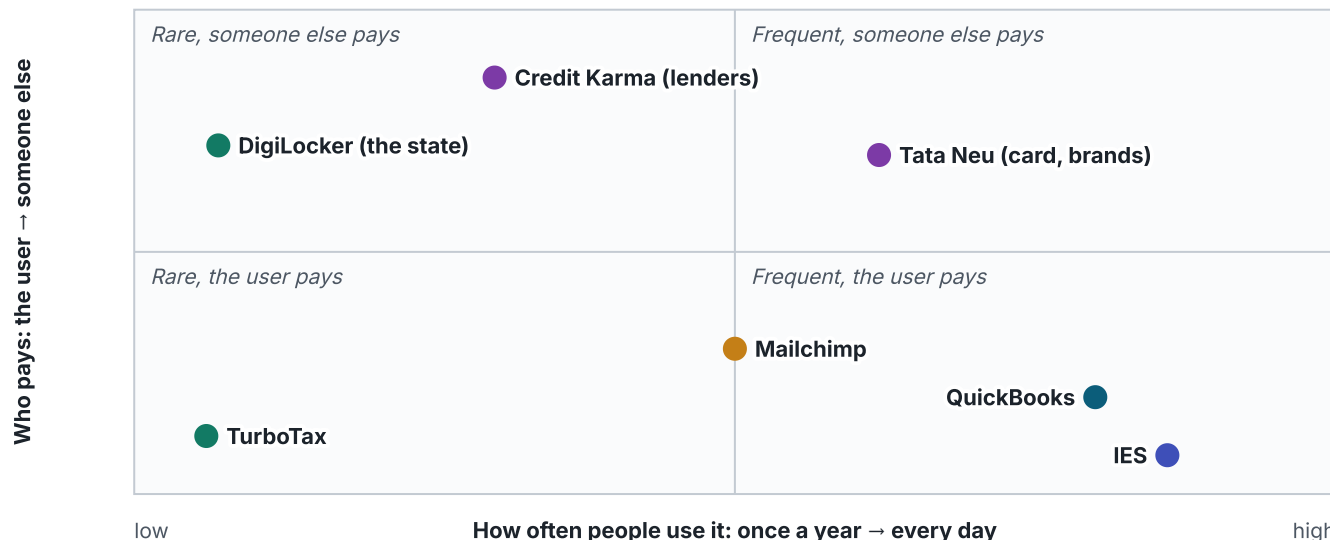


Figure 9.48 · How often it is used, and who pays. Products in the top row can be free because someone else pays; products on the left must earn a lot from each rare use. TurboTax's answer to rare use is selling expert help; Tata Neu's answer to weak frequency was the card. *Author's placement; positions are judgement, not measured values.*

IN THE INTERVIEW · HOW THESE PRODUCTS COME UP

- **“Improve TurboTax.”** It is used once a year and feared; start with the step where people drop out or pay for help, and remember trust is the product.
- **“Should QuickBooks re-enter India?”** Ask what the local rules and channel are (GST, Tally, accountants) and whether Intuit can win them; a global product alone lost last time.
- **“How does a free app like Credit Karma make money, and what can go wrong?”** Lenders pay per approval; the risk is ranking offers by fee instead of by the member's chances.
- **“Why did Tata Neu struggle, and what would you change?”** No daily reason to open it; build around what worked (the card) and make coins easy to spend everywhere.

Shipping without breaking things, and one system designed end to end

Every product in this sector reaches people through software on their phones: the Neu app, DigiLocker, QuickBooks, a voter app. Getting a change to crores of phones safely is its own craft. Four ideas cover it: how mobile releases work, feature flags (switches the team controls from its servers), staged roll-outs (giving a change to a few people first), and observability (seeing what the software is doing). It ends with a capstone: the design of IRCTC's Tatkal booking at 10 am, where caching, queues, consistency, scaling and payments all meet in one system. One running picture helps with the first four: a busy restaurant adding a new dish to the menu.

At the restaurant	The idea	What it means for software
Printed menus already handed out to diners can't be changed	Mobile releases	Once an app version is on a phone, you can't pull it back
A blackboard of today's specials, changed with chalk	Server-driven screens	The server decides what the app shows, without a new version
A switch in the kitchen: "stop serving the new biryani"	Feature flag , kill switch	Turn a feature off for everyone in seconds, no release
Serving the new dish at two tables first	Staged roll-out , canary	Give a change to 1% of users, watch, then widen
Staff tasting it before any customer does	Ring deployment	Staff first, then beta testers, then the public
The manager counting plates sent back each hour	Metrics	Numbers over time: crashes, errors, speed
The kitchen diary: "table 7, 8:14 pm, too salty"	Logs	A record of each event, with details
Following one order from ticket to table, minute by minute	Traces	One request's journey through every service, with timings
"No more than 1 in 1,000 plates sent back this month"	Error budget	How much failure the target allows before you slow down

A. Mobile releases: why shipping to phones is different

Everyday picture. A restaurant that changes a dish on its website can fix a typo in a minute. A restaurant that has printed 10,000 menus and handed them to customers can't; the old menus stay in circulation until people throw them away.

How it works. A website lives on the company's servers, so everyone gets the new version the moment it is shipped, and a bad change can be rolled back in minutes. A mobile app is a copy on every phone (Figure 9.49). A new version is built and signed, sent to Apple's App Store and Google Play for review (Apple's review usually takes hours to a day or two), and then reaches phones over days and weeks as they update, some automatically and some only when the user taps "update". Meanwhile the old versions keep running. Teams therefore ship on a **release train**: a new version every one or two weeks on a fixed day, whatever is ready gets on, and what isn't waits for the next train. An urgent fix (a **hotfix**) jumps the queue with its own release.

A website: one copy, on your servers



A mobile app: a copy on every phone

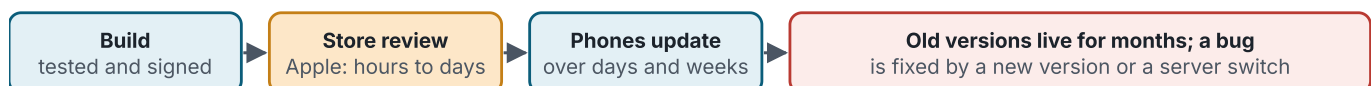


Figure 9.49 · A web release vs a mobile release. On the web you can undo a mistake in minutes. On phones you can only ship a newer version and wait, which is why mobile teams rely on switches they control from their servers. *Author's diagram; review times typical of 2025–26.*

Worked example. An app with 1 crore users ships version 5.2. After a week about 40% have it; after four weeks about 68%; after three months still only about 75% (Figure 9.50). So 25 lakh people are running code three months old, on many phone models and operating-system versions. Every server change must keep working for all of them: if the server stops sending a field the old version expects, that version breaks for 25 lakh people.

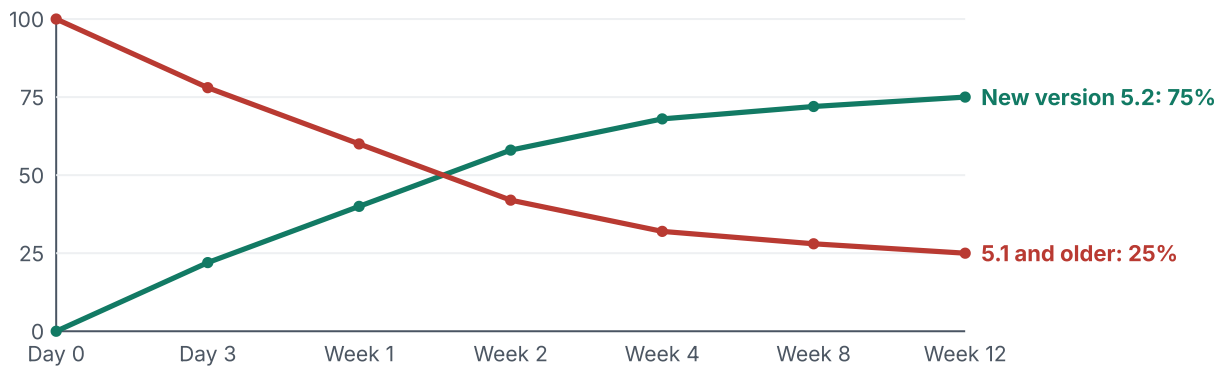
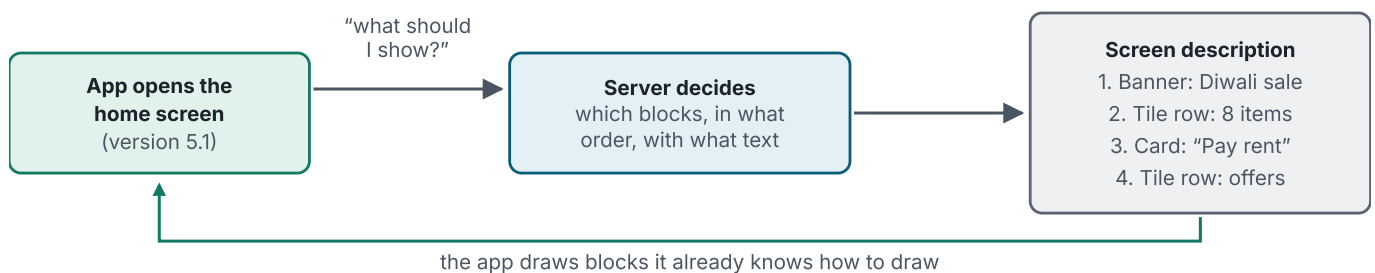


Figure 9.50 · How fast a new version spreads. Adoption is quick at first and then slows; a long tail of old versions stays for months, especially on older phones with little storage. *Illustrative; the shape is typical of Android apps in India.*

Two tools for the tail. A **minimum supported version** is a number the server checks: if a phone's app is older, it shows "please update to continue". It is used sparingly, for security fixes or when an old version would break (a payments app changing how it talks to NPCI), because every forced update loses some users who don't update. **Server-driven screens** go the other way: the app asks the server what to show, in what order, with what text, and draws blocks it already knows (Figure 9.51). A shopping app can rearrange its home screen for a Diwali sale on every version at once. The limit: a new kind of block the old app doesn't know still needs a release.



Change the order or the text today, on every version; a brand-new kind of block still needs a new release.

Figure 9.51 · A server-driven home screen. The server sends a description of the screen; the app draws it. Changing the order or the words needs no release; a new kind of block does. *Author's diagram; a pattern used widely by large consumer apps.*

Where it breaks. Apple or Google can reject a release a day before a big launch. A bug in an old version can't be fixed at all for users who never update. And the store rules themselves change: both stores require apps to support recent operating systems within set deadlines. **PM decisions:** what goes on this week's train and what waits; when a bug justifies a hot-fix; when to force an update and accept losing some users; which screens should be server-driven so that marketing doesn't need an engineer for every change.

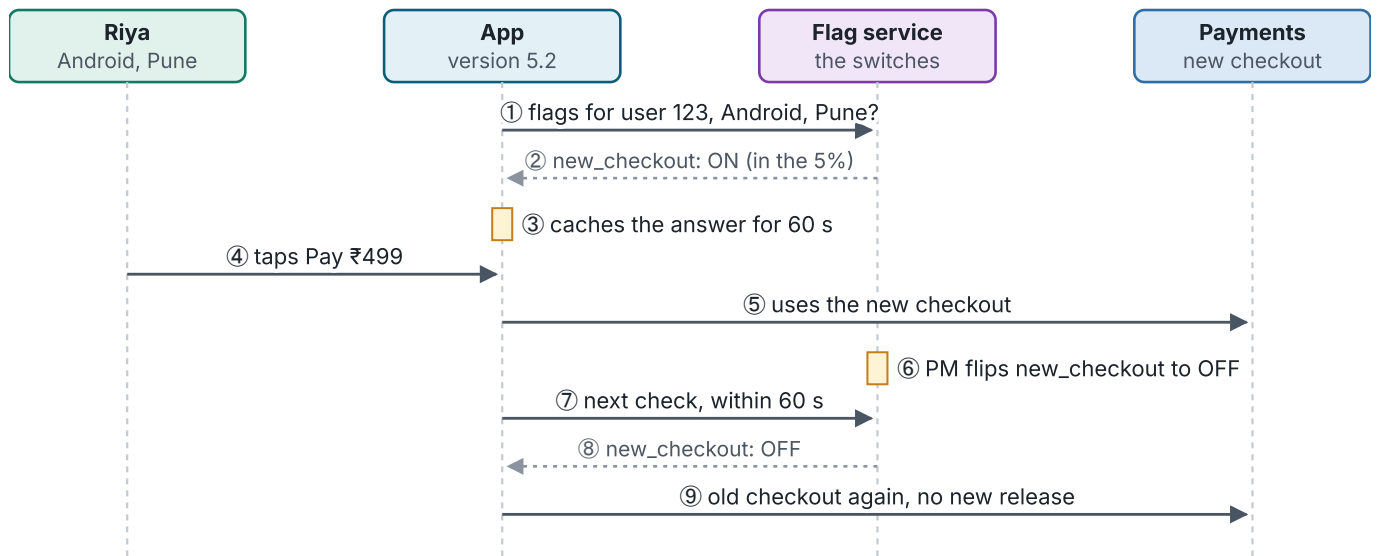
IN THE INTERVIEW · RELEASES

"Why can't we just roll back the app?" Because the old version is already on crores of phones; you ship a fix or switch the feature off from the server. "Should we force everyone to update?" Only for security or when the old version would break; each forced update loses some users, so measure how many.

B. Feature flags: switches the team controls from its servers

Everyday picture. The restaurant adds a new biryani. The chef keeps a switch on the kitchen wall. If diners send it back, the switch goes off and waiters stop offering it; nobody reprints the menu.

How it works. A **feature flag** is an if-statement in the code whose answer comes from a server: "if new_checkout is on for this user, show the new checkout; otherwise the old one." Both versions of the code are already inside the app. When the app starts, it asks a flag service which flags are on for this user, giving details such as the user's ID, phone type, city and app version, and keeps the answer for a short time (Figure 9.52). The team can then turn a feature on for 5% of Android users in Pune, for staff only, or for everyone, from a web page, in seconds. There are four common kinds. **Release flags** hide unfinished work until it is ready. **Experiment flags** split users into groups for an A/B test. **Operations flags**, or **kill switches**, turn a feature off in an emergency. **Permission flags** give a feature to a plan or a group of customers (Enterprise Suite features for IES customers only).



The code for both checkouts is already on the phone; the flag only chooses which one runs.

Figure 9.52 · A feature flag at work. The app asks which switches are on (①–②), remembers for 60 seconds (③), and picks up a change at its next check (⑦–⑨). How often it checks decides how fast a kill switch works. *Author's diagram; timings illustrative.*

Worked example. A payments app switches its new UPI checkout on for 20% of users at 10:04 am. Failures rise from the usual 1.2% to 3.8% of payments. The alert fires at 10:07; the PM and the on-call engineer flip the flag off at 10:08:30; within a minute, as phones check in again, failures fall back to 1.2% (Figure 9.53). About 4.5 minutes at 20% of traffic went through the bad checkout. If the app handles 3,000 payments a minute, that is $3,000 \times 20\% \times 4.5 \times (3.8\% - 1.2\%) \approx 70$ extra failed payments, not the thousands a new app release would have taken hours to stop.

Payment failures (%) minute by minute after the new checkout is switched on for 20% of users (illustrative)



Figure 9.53 · A kill switch, minute by minute. The damage is set by two delays: how fast the alert fires, and how often phones check their flags. Both are design choices. *Illustrative numbers.*

Where it breaks. Flags pile up. An old flag nobody removes becomes a trap: in August 2012 the US trading firm Knight Capital reused an old flag for new code, one of its eight servers still ran the old code the flag switched on, and it lost \$440 million in 45 minutes. Too many flags also multiply the combinations nobody has tested (ten on-off flags make 1,024 versions of the app), and a flag service that goes down must fail safe, falling back to the old behaviour. **PM decisions:** which features ship behind a flag (anything touching money, sign-up or a big screen); who may flip it and how fast phones must notice; the rule that every flag has an owner and a removal date, so flag debt is cleared each quarter.

IN THE INTERVIEW · FEATURE FLAGS

“How would you launch a risky change to checkout?” Behind a flag, to staff first, then 1%, then wider, with payment success as the guardrail and a kill switch that works in under a minute. “What’s the risk of flags?” Old flags nobody removes; set an owner and a removal date for each.

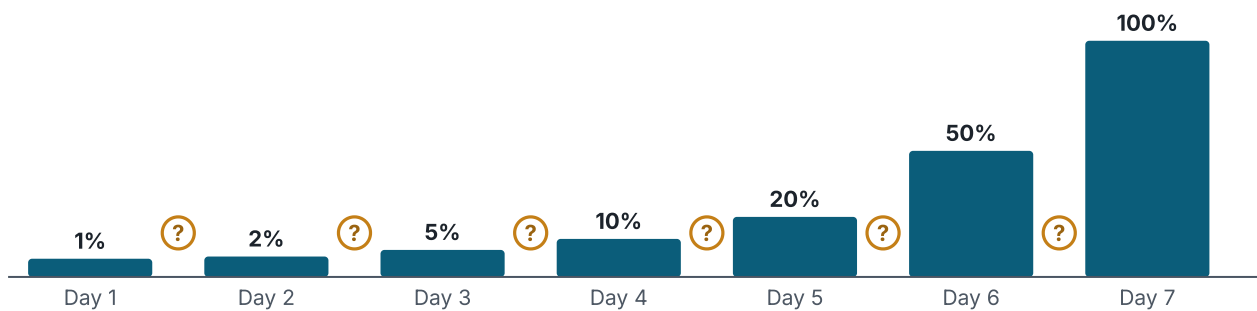
C. Staged roll-outs: a few people first

Everyday picture. Before the new biryani goes on every table, the chef serves it to the staff at lunch, then to two regular tables on a quiet Tuesday. If it comes back uneaten, only a few people were affected, and the chef learns why before Saturday night.

How it works. A **staged roll-out** gives a new version or feature to a small share of users first and widens it only if the numbers stay healthy. Both app stores support it. Apple’s **phased release** sends an update to automatic-update users over seven days: 1%, 2%, 5%, 10%, 20%, 50%, then everyone, and the team can pause it (Figure 9.54). Google Play’s **staged roll-out** lets the team pick the percentage and halt at any time. Server changes and flags use the same idea: a **canary** (named after the birds miners took underground to warn of gas) is the small group that gets the change first. Many firms go in **rings**: staff, then beta testers who signed up for early versions, then 1% of the public, then everyone. At each step the

team compares the new group with everyone else on a few numbers agreed in advance: crash-free sessions, payment success, app-store ratings, support tickets.

Apple's 7-day phased release: share of users who get the update (a gate before each step)



At each gate (?): crash-free sessions, payment success, ratings and support tickets vs the old version. Any one fails: pause.

Figure 9.54 · A seven-day phased release, with a gate before each step. Each step widens only if the checks pass. Most problems show at 1–5%, when few people have been hurt. *Apple App Store Connect phased release (percentages as documented); gate checks are the author's example.*

Worked example. An app with 1 crore users has 3 crore sessions a day. On the old version, 99.8% of sessions don't crash: 0.2%, or 60,000 crashes a day. On day 3 of a roll-out the new version is at 5%, with 15 lakh sessions a day, and it is crash-free on only 99.1% of them: 13,500 crashes where 3,000 were expected. That looks small in absolute terms, but at 100% the same rate means 2.7 lakh crashes a day, 2.1 lakh more than today (Figure 9.55). The rule agreed before the roll-out ("pause if crash-free sessions fall by more than 0.2 points") makes the call easy: pause, fix, resume.

Crashes a day for an app with 1 crore users and 3 crore sessions (illustrative)

Old version, all users	60,000 0.2% of 3 crore sessions a day crash
New version, if at 5%	13,500 0.9% of its 15 lakh sessions (seen on day 3)
New version, if at 100%	270,000 0.9% of 3 crore: 2.1 lakh extra crashes a day

Figure 9.55 · Why a small group tells you about everyone. A crash rate seen at 5% is the rate you will get at 100%; only the number of people hurt changes. *Illustrative numbers.*

Where it breaks. Some problems only appear at full scale: a server that copes with 5% of traffic falls over at 100%. Small groups can be unlike everyone else (staff have new phones and good Wi-Fi), so bugs on cheap phones on slow networks slip through. And a change that skips the stages can hurt everyone at once. On 19 July 2024, a faulty update from the security company CrowdStrike went to all its Windows customers together and crashed about 85 lakh computers worldwide (Microsoft's estimate), grounding flights and disrupting banks and hospitals, including in India. A roll-out to 1% first would have caught it in minutes. **PM decisions:** the gate numbers and the pause rule, set before the roll-out starts; how long to wait at each step (a day for a big release, an hour for a small change); which users go first (staff and beta testers, not the biggest business customers); and the rule that "just a configuration change" goes through stages too.

IN THE INTERVIEW · STAGED ROLL-OUTS

"Our last release crashed on many phones. What would you change?" Stage every release (staff, 1%, 5%, and on), agree the pause rule in advance, and make sure the early groups include cheap phones and slow networks, where Indian users are.

D. Observability: seeing what the software is doing

Everyday picture. A courier company knows three kinds of things about its parcels (Figure 9.56). A dashboard shows counts (12,400 delivered this hour, 3% late). Each hub keeps a diary ("14:02, parcel 8841 scanned, label torn"). And any single parcel can be tracked hub by hub, with the time it spent at each.

Metrics counts over time Courier company: "12,400 parcels delivered this hour; 3% late" Is something wrong?	Logs a diary of events Courier company: "Hub 7, 14:02: parcel 8841 scanned, label torn" What exactly happened?	Traces one request's journey Courier company: "Parcel 8841: Pune 2 h, Nagpur 19 h, Delhi 3 h" Where did the time go?
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Figure 9.56 · Metrics, logs and traces, as a courier company. Metrics tell you something is wrong; logs tell you what happened; traces tell you where the time went. *Author's analogy.*

How it works. **Metrics** are numbers counted over time: requests a second, error rate, crash-free sessions, how long a screen takes to load. They are cheap to store and good for alerts. **Logs** are records of single events with details (which user, which bank, which error code); they are what an engineer reads to find the cause. **Traces** follow one request through every service it touches, with the time spent in each. On phones, **crash reporting** tools collect every crash with

the phone model and the line of code, and **real-user monitoring** measures what users actually experience: how long the app takes to open, how long each screen takes to load, reported as the median and the 95th percentile (the time within which 95 of 100 loads finish, the number the slowest users feel).

Worked example: a slow payment screen. Users complain the payment screen hangs. The metric shows the screen's 95th-percentile load time rose from 1.8 to 4 seconds this week. A trace of one slow request shows where the 4 seconds go (Figure 9.57): 0.1 s to send, 0.2 s to check the login, 0.3 s to fetch offers, 3.2 s waiting for one bank's API to confirm, 0.2 s to draw the screen. The logs for that bank show a timeout error on its side since Tuesday. So the fix is not in the app: show a "confirming with your bank" state after 1.5 seconds, send users to other banks' routes if that bank is slow, and raise it with the bank.

One slow payment screen, as a trace: 4.0 seconds in total (illustrative)

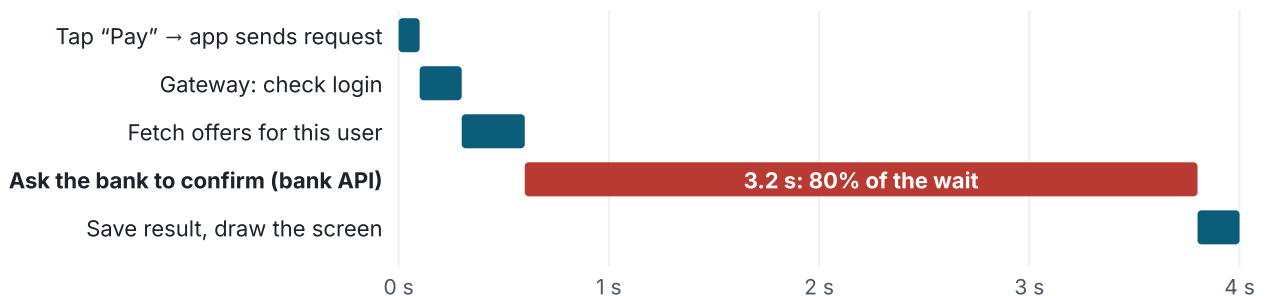


Figure 9.57 · A trace of one slow payment screen. Four of five steps are fast; one bank's API takes 80% of the time. Without a trace, a team might spend a week speeding up the app's own code. *Illustrative timings.*

Alerts that mean something. An alert on every error would wake the on-call engineer all night. Teams instead set a target, such as 99.9% of payment screens loading within 2 seconds each month, which leaves an **error budget** of 43 minutes of slow screens a month. They alert when the budget is being used up too fast: if a day uses a fortnight's worth, someone is paged (Figure 9.58). The same budget helps decide releases: with plenty left, ship; nearly used up, freeze risky changes and fix reliability first.

Error budget: 43 minutes of slow payment screens allowed in a 30-day month (99.9% target)

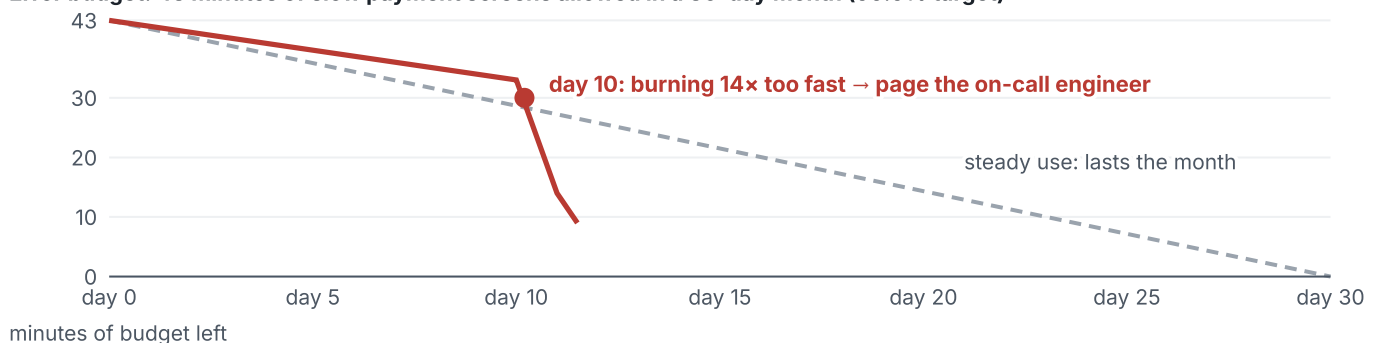


Figure 9.58 · Alerting on the burn rate. At a steady pace the budget lasts the month (grey). On day 10 a bank problem burns it 14 times too fast, and that is the moment to page someone, not every single error. *Illustrative; 99.9% of a 30-day month = 43 minutes.*

Where it breaks. Collecting everything costs money: logs from crores of users can cost more than the servers they describe, so teams sample. Logs can leak personal data (a full card number or Aadhaar number written into a log is a breach under the DPDP Act). Averages hide pain: a median of 1 second can sit next to a 95th percentile of 8 seconds for users on slow networks. And dashboards nobody watches are decoration. **PM decisions:** which few numbers define "working" for each feature (the ones the gates and alerts use); the targets and budgets; what may never be logged; and a rule that every new feature ships with its metrics and alerts, not a month later.

IN THE INTERVIEW · OBSERVABILITY

"Users say the app is slow. What do you do?" Find which screen and which users (95th percentile by phone and network), trace a slow request to see where the time goes, then fix the slowest step, which is often another company's API, not your code. "What's an error budget?" The failure your target allows; when it runs low, slow down releases.

E. Capstone: designing Tatkal booking at 10 am, at PM level

Everyday picture. A famous temple on a festival morning. Lakhs of devotees arrive when the gates open at 5 am, but the inner hall holds a few hundred at a time. A well-run temple doesn't let everyone rush the door. It sets up a queue outside with barriers, gives each person a token with a time, lets a fixed number in each minute, checks tickets at the gate so touts can't jump the line, keeps a separate fast lane for staff, and tells people the true wait. The prasad counter shuts for an hour if the hall gets crowded; the darshan never stops.

The problem. Tatkal tickets for the next day open at 10 am for AC classes and 11 am for sleeper and other non-AC classes. Most of the berths on busy routes go in the first few minutes, and lakhs of people, plus agents and bots, try at once. Traffic that is calm at 9:58 is many times higher at 10:00:00 (Figure 9.59). The system must never sell the same berth twice, must not charge anyone without issuing a ticket, and must be seen as fair. IRCTC booked a record 20,06,353 tickets on 7 September 2026, after roughly doubling its capacity with new servers and software changes and adding anti-bot and CDN protection.

Requests a second at the Tatkal booking system around 10 am (thousands, illustrative)

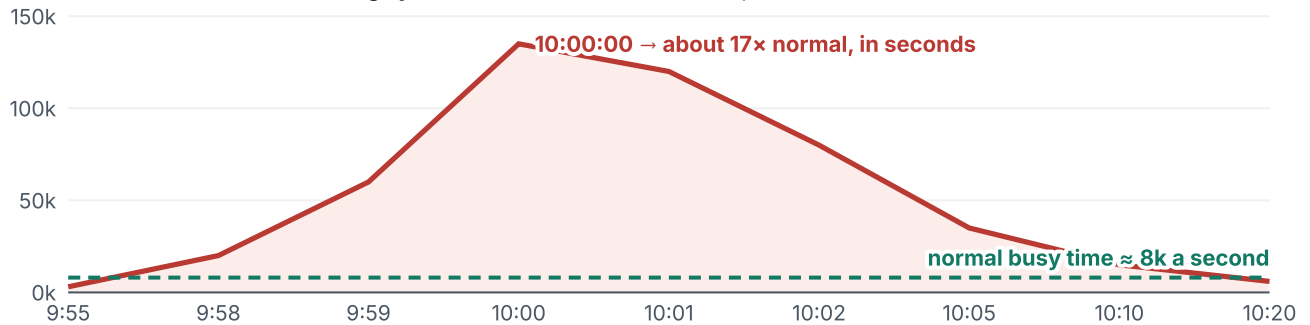
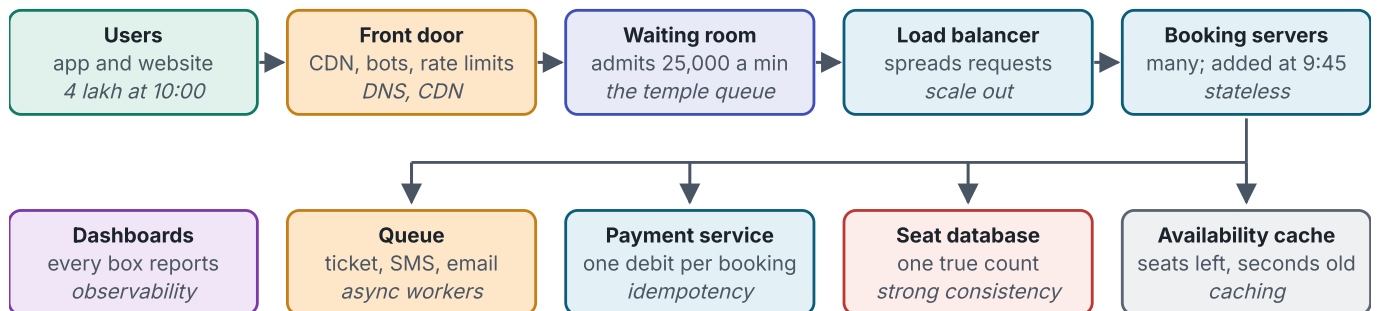


Figure 9.59 · The 10 am wall. Demand jumps about 17 times in seconds and fades within 20 minutes. Adding servers after the jump is too late; capacity must be in place by 9:45. *Illustrative shape and numbers; IRCTC's record of 20.06 lakh tickets in a day (7 Sep 2026, Business Today).*

Requirements, with numbers. Say 4 lakh people are logged in at 10:00 for Tatkal, each refreshing the seat count every three seconds: $4,00,000 \div 3 \approx 1.3$ lakh requests a second, against about 8,000 in a normal busy minute. Say the system can safely complete 25,000 bookings a minute. Then the job is to turn a crowd of 4 lakh into an orderly flow of 25,000 a minute, while keeping the seat count exact. All numbers here are illustrative.

The design, piece by piece

Figure 9.60 puts it together; each piece is explained in a few lines.

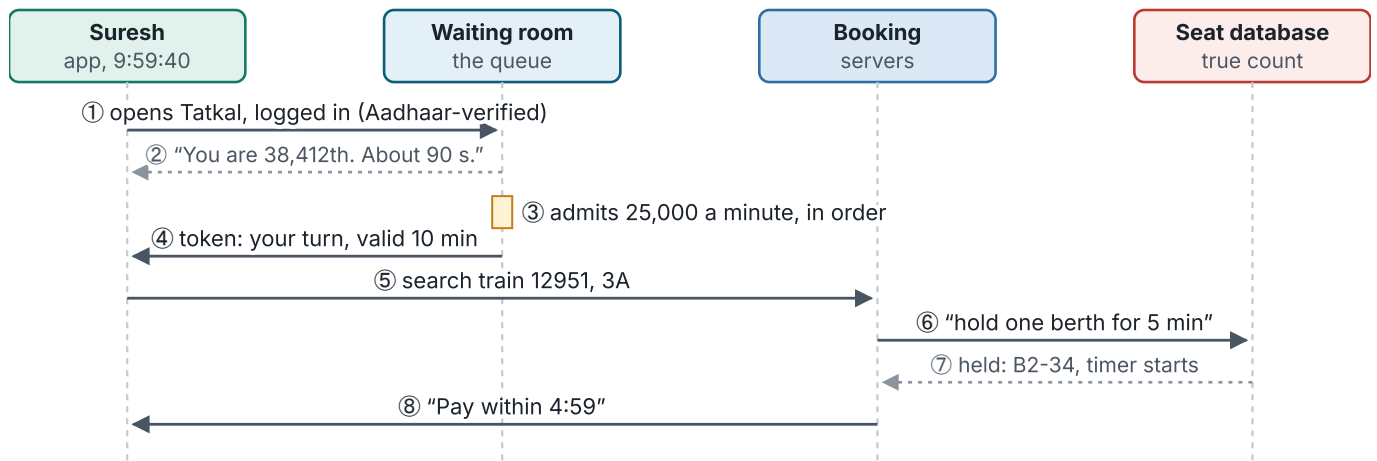


Top row: controlling the crowd. Bottom row: keeping the count exact and the money safe. Italics: the mechanism at work.

Figure 9.60 · Tatkal booking, the whole system. The top row controls the crowd; the bottom row keeps the seat count exact and the money safe. *Author's design at PM level, not IRCTC's actual architecture.*

1. The front door (internet basics, CDNs). DNS sends users to the nearest entry point; a CDN serves the static parts of the site (pages, images, train lists) so they never touch the booking servers. A bot filter and **rate limits** (a cap on requests per account and per network address each second) turn away scripts. Since 1 July 2025 only Aadhaar-authenticated users can book Tatkal online, with an Aadhaar OTP from 15 July 2025, and authorised agents can't book in the first 30 minutes of each window. Those are product rules doing a security job: they make each bot cost a real identity.

2. The virtual waiting room. Like the temple queue, it lets a fixed number of people through each minute, in order of arrival, and tells everyone else where they stand (Figure 9.61). A user who is 38,412th sees "about 90 seconds", not a spinning wheel. The waiting room is cheap to run because a waiting user only asks "is it my turn?" every few seconds. The admitted user gets a token valid for 10 minutes, so the booking servers see only as many people as they can serve.



The queue protects the database: only as many people reach booking as it can serve.

Figure 9.61 · Inside the waiting room. The queue absorbs the crowd (①–③); only admitted users reach booking (④–⑤); a berth is held while they pay (⑥–⑧). Author's diagram; numbers illustrative.

3. Scaling before the wall (scaling and reliability). Booking servers keep no user data of their own (they are stateless), so more can be added behind the load balancer. Because the spike is known to the minute, they are added on a schedule at 9:45, not by an autoscaler that reacts after the rush has begun.

4. Two kinds of seat count (data, caching, consistency). The number shown while browsing ("3A: available 42") comes from a cache and may be a few seconds old; it is labelled as approximate. The count that matters, the one used to hold a berth, comes from the seat database, where each train has one authoritative record and only one booking can take a given berth. Browsing can be slightly stale; booking must be exact.

5. Holding a berth while the user pays. When a user picks a berth, it is **held** for five minutes: nobody else can take it, and the user sees a countdown (Figure 9.62). If payment succeeds, it becomes booked; if the timer runs out, it goes back on sale. The number of berths held at any moment is the rate of new holds times the hold time: 25,000 a minute $\times$ 5 minutes $\approx$ 1.25 lakh berths. If 30% of holders never pay, those berths return to sale minutes later, to someone further back in the queue. A shorter timer returns them sooner but fails people on slow UPI apps; this is a product choice, not a technical one.

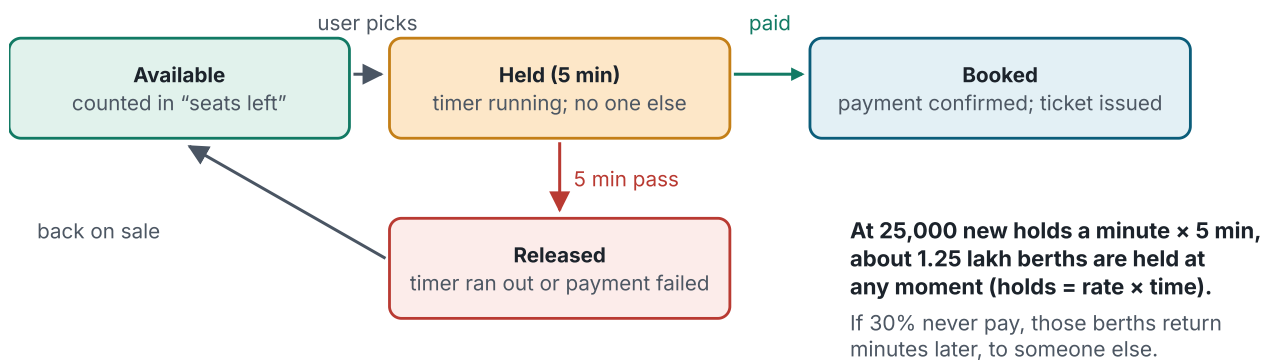
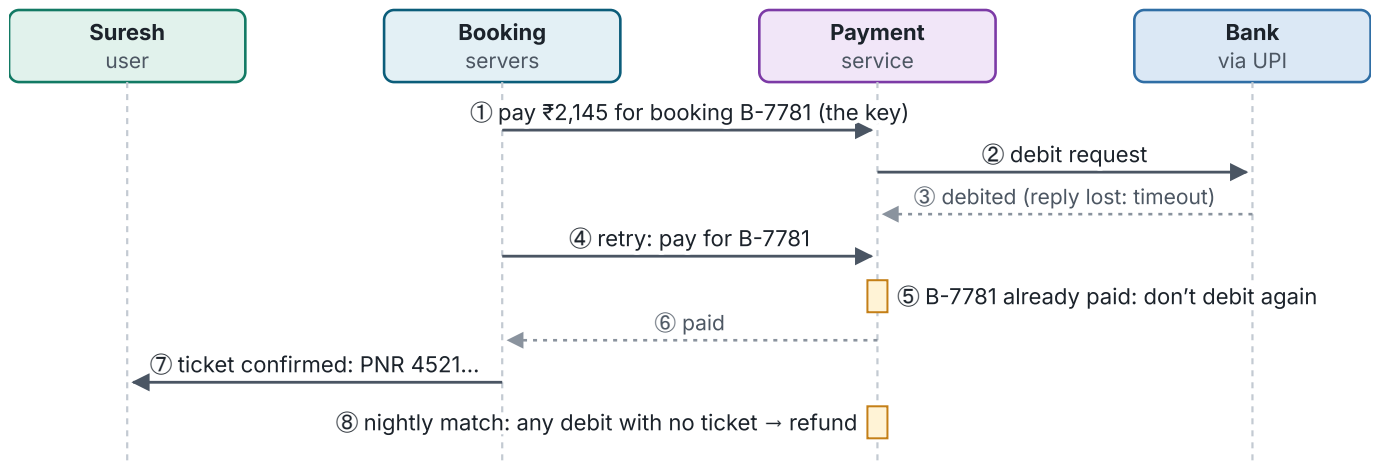


Figure 9.62 · The life of one berth. A hold is a promise with a timer. Its length decides how many berths are locked up at any moment and how fairly they come back. Author's diagram; numbers illustrative.

6. Payment that can't double-charge (APIs, idempotency). Every payment request carries the booking ID. If the bank debits the money but the reply is lost, the retry arrives with the same ID and the payment service sees it is already paid (Figure 9.63). The case users fear most, "money gone, no ticket", is caught by a nightly match of every debit against every ticket, with an automatic refund for any debit that has no ticket.

7. Everything after the booking goes in a queue (queues, notifications). The PDF ticket, the SMS and the email are put in a queue and sent by separate workers a few seconds later, so a slow SMS provider never holds up the next booking.



The booking ID travels with every retry, so a lost reply can't turn into a second debit.

Figure 9.63 · A lost reply that doesn't become a second debit. The booking ID (the key) makes the retry safe (④–⑤); the nightly match catches anything that slipped through (⑧). Author's diagram; amount illustrative.

8. A degradation ladder for the morning (graceful degradation). Decided in advance and rehearsed: switch off meal and hotel offers at 9:45, serve fare enquiries and PNR checks from copies, pause new sign-ups, and never touch holding, paying and ticketing (Figure 9.64).

What gives way first on a Tatkal morning (top) and what never does (bottom): a degradation ladder

Meals, hotel, tour offers	switched off at 9:45, back at 11:30
Fare enquiry for other quotas	shown from a copy made at 9:30
PNR status checks	answered from a cache, up to 1 min old
New sign-ups	paused 9:55–10:30; log-in stays open
Seat availability	cached, a few seconds old, marked "approx."
Hold a berth, pay, get the ticket	never switched off; strict count

Figure 9.64 · What gives way first. Each rung frees capacity for the one job that must not fail. The PM decides the order; engineers build the switches. Author's example ladder.

9. Watching the morning (observability). One screen for 9:45 to 10:30: people waiting and admitted per minute; holds, bookings and releases per minute; payment success by bank; 95th-percentile time from "admitted" to "ticket"; error budget left; bot blocks. Each has an owner and a threshold that triggers a named action.

The PM trade-offs

Decision	Option A	Option B
Order in the queue	First come, first served: feels fair, rewards the fastest connection	Random order among everyone who arrived by 10:00: fairer to slow phones, feels like a lottery
Stopping bots	Aadhaar OTP for every Tatkal booking: few bots	Lighter checks: fewer honest people shut out (lost phone, Aadhaar-number mismatch)
Hold time	3 minutes: berths return faster	7 minutes: fewer failures for people on slow payment apps
What to show the 40,000th person	"About 90 seconds" plus a live count of berths left, even when it is falling fast	A spinner: calmer, but people refresh and add load
When seats run out	Tell people in the queue as soon as a train is full	Let them reach the booking page: costly and cruel

KEY IDEA · PROTECT THE ONE JOB, AND BE HONEST WITH THE CROWD

Tatkal is a small, exact job (hold a berth, take the money, issue a ticket) surrounded by a crowd. The design controls the crowd first (front door, waiting room, rate limits), keeps the exact job strict (one seat record, holds with timers, one payment per booking), and gives up everything else under load. The most useful thing a PM adds is honesty: a queue position, a wait time, a countdown and an early "sold out" turn a stampede into a line.

IN THE INTERVIEW · SYSTEM DESIGN AT PM LEVEL

"Design Tatkal booking" (or a flash sale, IPL tickets, a vaccine slot opening). Start with the numbers (people, requests a second, what can be completed a minute), then the crowd controls (front door, waiting room, rate limits), then the exact part (one seat record, holds, idempotent payment), then what gives way under load, then what you watch. End with the trade-offs you, the PM, must decide: queue order, bots vs honest users, hold time, what to show people who won't get a seat.

Connect the dots

A speaker, a loyalty coin, a voter app and a tax return all earn their keep after the first transaction. The cases, the products and the technology kept returning to the same few ideas: find the moment when people act and make that moment easy; count what a unit really costs over its whole life; measure what you caused against a group that got nothing; and change live products in small, reversible steps.

Idea	In the cases	In the products	In the technology
Act at the moment	A plan-making prompt three days before polling; the booth in one tap	TurboTax pulls in the W-2; DigiLocker shows the licence at the checkpoint	A waiting room that tells the 40,000th person the wait
Count the whole life of a unit	A speaker's price less tax, shop and cost, plus services; a coin's cost after breakage	QuickBooks earns more from payments and payroll than from the plan	Holds = rate × time; crashes at 5% predict crashes at 100%
Measure what you caused	A random holdout for voter reminders and for the loyalty bonus	Credit Karma's "very good" approval rate; Intuit's days-faster payments	Staged roll-outs compare the new group with everyone else
Price for the buyer who can buy	iPhone homes, not all homes; earn rates set by each brand's margin	TurboTax's ladder of help; IES priced by contract	An error budget sets how much risk a release can take
Small, reversible steps	Four features before polling day, the rest later	Neu refocused on the card that worked	Flags, kill switches, phased releases, a degradation ladder

BACK TO LUCKNOW: A SKETCH ANSWER TO THE COLD OPEN

"Ananya will spend ₹25,000 on gadgets but may not spend 20 minutes voting. What would you build?" The gadgets won because Apple made each step easy and rewarding at the moment of purchase; voting asks her to register by a deadline, find a booth and set aside a morning, with no prompt at any of those moments. So don't build a feature-rich voter app. Build the three moments. Four weeks before the registration deadline, a message: "Check you're on the list" with a one-tap search. Three days before polling day: "Booth 14, Government School, open 7 to 6. What time will you go?", saved to her calendar. On the morning, a reminder at the time she chose. Put the candidates' own sworn declarations side by side, the same for everyone, and nothing that looks like a score.

"How would you know it worked?" Hold out a random group of young voters who get no messages, and compare turnout between the two groups on the official record. A 2–4 point lift among the people reached is a good result; claiming the city's 55% will become 70% is not. Ship it the way good teams ship software: a small group first, with a pause rule (complaints of bias, wrong booth details), and switches that can stop any message in seconds.

Numbers worth remembering

Number	Value	Source and date
Lok Sabha turnout	67.4% → 65.8%	ECI, 2019 and 2024
Turnout lift from a plan-making call; from a "neighbours will see" mailer	+4.1 points; +8.1 points	Nickerson and Rogers (2010); Gerber, Green and Larimer (2008)
HomePod mini and HomePod, India list prices	₹15,900; ₹44,900	Apple India store, Sep 2026
Apple services revenue	about \$109 bn	Apple, FY2025
Tata Digital revenue and loss	₹35,990 Cr; ₹4,974 Cr	Filings via Inc42, FY26
Tata Neu HDFC Bank cards issued	20 lakh+	Business Standard, Mar 2025
DigiLocker users; documents issued	67.6 crore; 950 crore+	PIB, Mar 2026
Intuit revenue; TurboTax; Credit Karma	\$21.4 bn; \$5.3 bn; \$2.6 bn	Intuit, FY2026 (to 31 Jul 2026)
Swiggy food orders in a quarter; monthly food users	11.45 crore; 1.92 crore	Swiggy, Apr–Jun 2026
IRCTC record tickets in a day	20,06,353	7 Sep 2026
Apple phased release steps	1, 2, 5, 10, 20, 50, 100% over 7 days	App Store Connect
CrowdStrike update that went to everyone at once	about 85 lakh PCs crashed	Microsoft estimate, 19 Jul 2024